

Adaptive fault diagnosis in power transmission lines using deep learning and LSTM autoencoders for enhancing grid reliability

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ABSTRACT

The increasing complexity and dynamic nature of modern electrical grids necessitate advanced, adaptive fault diagnosis systems to maintain high reliability and ensure minimal downtime. This study presents a novel, adaptive fault detection and localization method for three-phase transmission lines utilizing a Long Short-Term Memory (LSTM) Autoencoder. The model operates in an unsupervised manner, learning the standard operational patterns from three-phase voltage and current signals and identifying faults as anomalies through high reconstruction errors. Trained and tested on a comprehensive dataset of over 50,000 simulated fault events generated in MATLAB/Simulink and rigorously validated on 1,000 real-world fault instances from an open-source repository, the proposed method demonstrates exceptional performance and robustness. It achieves a 98 % accuracy and a 2 % false positive rate, outperforming traditional methods (DFT, Wavelet) and other deep learning benchmarks (standard LSTM, 1D-CNN) by over 15 % in F1-score. The model exhibits strong resilience to noise, maintaining an F1-score above 92 % at a 20 dB SNR, and demonstrates computational efficiency suitable for real-time deployment. These results validate the LSTM Autoencoder as a potent and practically significant tool for enhancing adaptive fault management and real-time monitoring in modern power systems, directly contributing to improved grid reliability and operational efficiency.

1. Introduction

Power transmission systems are crucial for the reliable delivery of electrical energy from generation sources to end-users across vast distances. These systems form the backbone of modern electrical grids, ensuring that electricity generated at power plants is efficiently transmitted to homes, industries, and businesses. Maintaining the stability and efficiency of these systems is vital for preventing outages and ensuring uninterrupted power supply [1]. Effective fault detection and localization are essential for the reliability and efficiency of power transmission networks. Faults in these systems, such as short circuits, line interruptions, or equipment failures, can lead to significant disruptions in service, potential damage to infrastructure, and increased maintenance costs. Timely and accurate identification of faults helps minimize downtime, prevent damage, and optimize repair processes [2]. Traditional fault detection methods have relied on impedance-

based techniques, relay settings, and other conventional approaches. These methods involve measuring the impedance of transmission lines to detect faults and using relays to trigger protective actions [3]. While these approaches have been effective to some extent, they face limitations in handling the increasing complexity and dynamic nature of modern power grids. They often struggle to provide real-time responses and accurate fault localization in complex grid scenarios [4]. Recent advancements in deep learning have introduced sophisticated techniques that enhance fault detection and localization capabilities [5]. Deep learning, particularly through deep signal processing, utilizes advanced neural networks to analyze complex signal patterns and temporal dependencies. These methods have demonstrated superior performance compared to traditional techniques by offering improved accuracy and more detailed insights into fault characteristics and locations [6]. Deep signal processing techniques, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM)

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networks, are particularly promising for power system applications. CNNs excel at spatial feature extraction, while LSTMs are adept at capturing temporal dependencies in time-series data [7]. When combined, these techniques can analyze voltage and current waveforms with high precision, enabling more accurate detection and localization of faults [8].

Among the deep learning methods, LSTM autoencoders have emerged as a powerful tool for fault detection and localization. LSTM networks are designed to handle sequential data and capture long-term dependencies, making them ideal for analyzing time-series data from power systems [9]. Autoencoders, on the other hand, are effective for anomaly detection by reconstructing data and identifying deviations from normal patterns. The integration of LSTM and autoencoder techniques provides enhanced fault detection capabilities by improving accuracy and reducing false alarms [10]. The application of LSTM autoencoders in power transmission systems represents a significant advancement over traditional fault detection methods. These models can learn from historical fault data and detect anomalies with greater precision [11]. By analyzing complex signal patterns and identifying deviations, LSTM autoencoders offer more accurate fault classification and localization, leading to faster and more effective responses. Despite the advantages of deep learning techniques, several challenges remain in their practical deployment [12]. First, the introduction of an LSTM autoencoder pipeline that separates detection and localization while learning exclusively from normal data. Second, it provides a principled, percentile-based threshold calibration tied to an explicit false-alarm budget, replacing ad-hoc static settings. Third, it presents a comparative evaluation against a tuned threshold baseline and additional learners, reporting latency, AUROC and false-positive rate, and we analyze computational feasibility for deployment beyond MATLAB simulations. One major challenge is data variability, as real-world power systems exhibit diverse operating conditions and fault scenarios. This variability can impact the performance of deep learning models, requiring robust training and validation processes to ensure reliability across different conditions. Computational demands are another critical challenge [13]. Deep learning models, especially those involving LSTM autoencoders, require significant computational resources for training and real-time processing. Efficient algorithms and hardware are necessary to handle these demands and ensure timely fault detection and localization. Research efforts must focus on optimizing models and algorithms to balance accuracy with computational efficiency [14]. Integrating deep learning methods with existing power transmission infrastructure also presents challenges. Traditional systems are often based on older technologies and standards, making it necessary to develop methods for effectively incorporating advanced models without disrupting existing operations. This integration involves addressing compatibility issues and adapting models to work within the constraints of current infrastructure [15]. Ongoing research is crucial to address these challenges and optimize deep learning techniques for practical applications. This includes refining model architectures, improving training methodologies, and developing strategies for real-time fault detection and localization. By advancing these techniques, researchers aim to enhance the reliability and efficiency of power transmission systems [16]. The integration of emerging technologies, such as micro-Phasor Measurement Units (PMUs) and neural architecture search (NAS) algorithms, offers additional opportunities for improving fault detection and localization. Micro-PMUs provide real-time phasor data that enhances accuracy, while NAS algorithms optimize deep Transformer models for better performance. These technologies complement deep learning approaches and address issues related to scalability and real-time applicability. Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) are also contributing to advancements in fault detection and localization [17]. VAEs augment real-world data with synthetic examples, improving model training and accuracy, while GANs use probabilistic modeling to differentiate between normal and faulty behaviors. These approaches enhance the robustness of fault

detection systems. Graph Neural Networks (GNNs) further enhance fault localization by representing power systems as networks of nodes and edges [18]. GNNs enable precise identification of fault locations by analyzing the structural relationships within the power grid. This capability is particularly valuable for complex power systems with intricate network configurations [19].

1.1. Literature review

Deep learning techniques, particularly Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have demonstrated superior performance in transmission line fault classification. These methods offer high accuracy and reliability in handling complex datasets [20]. However, challenges persist in developing scalable models for real-time applications, especially in computationally constrained environments [21]. LSTM-based models are effective in detecting, diagnosing, and locating faults in large-scale power systems. Their ability to capture temporal dependencies and handle complex data sequences is well-documented. Despite their advantages, optimizing LSTM models for real-time fault detection remains an area needing further research [22]. Hybrid CNN-LSTM approaches have been successful in identifying fault types and locations by combining CNN's spatial feature extraction with LSTM's temporal sequence modeling [23].

Additionally, advancements such as CNN-based capsule networks and LSTM autoencoders have shown promise in enhancing fault detection accuracy and system robustness [24]. However, real-time implementation and scalability issues are still prevalent [25]. Recent research highlights the use of micro-Phasor Measurement Units (PMUs) for fault location, leveraging real-time phasor data to improve accuracy [26]. Neural architecture search (NAS) algorithms have also optimized deep Transformer models, enhancing fault detection capabilities [27]. These emerging technologies face challenges related to scalability and real-time applicability. Variational Autoencoders (VAEs) combined with machine learning algorithms have improved fault classification and localization by augmenting real-world data with synthetic examples [28]. This approach enhances model training and accuracy but requires further exploration of scalability and real-time performance. Deep signal anomaly detectors, including LSTM autoencoders, CNNs, RNNs, GANs, and Graph Neural Networks (GNNs), play a critical role in fault detection and localization in three-phase power systems. These models excel in analyzing complex time-series data and identifying abnormal patterns [29]. Among these, LSTM autoencoders are noted for their effectiveness in capturing temporal dependencies and performing real-time monitoring [30]. Recent advancements in deep learning have introduced CNN-based capsule networks for fault location in power distribution grids, providing improved accuracy in identifying and locating faults. These models outperform traditional CNNs by preserving spatial relationships and handling complex grid patterns more effectively [31]. The application of capsule networks in real-time fault detection is underexplored, particularly in terms of computational efficiency and adaptability to large-scale distribution grids. Optimized machine learning algorithms have shown significant promise in fault detection, classification, and localization along power grid lines. Techniques such as Support Vector Machines (SVM), decision trees, and neural networks have improved fault identification accuracy and response times. Despite these advancements, there is limited research on optimizing these algorithms for real-time, large-scale grid environments, particularly in enhancing computational efficiency and reducing false positives. Recent work on DC fault protection in meshed HVDC grids employs explicit denoising or robust feature extraction before classification to withstand measurement disturbances and switching noise [32]. Such schemes can be highly effective in DC environments with distinct transient signatures but often introduce an additional processing stage that must be tuned to network and device characteristics. In contrast, our method uses noise-aware training of an LSTM autoencoder to learn invariant

reconstructions of nominal AC behavior, thereby separating genuine grid anomalies from measurement fluctuations without a dedicated denoiser. This design choice reduces pipeline complexity and preserves low latency while retaining robustness, as evidenced by our $< 1\%$ false positive rate under noise augmentation. A more extensive head-to-head comparison with explicit denoising in AC transmission remains promising future work.

Hybrid CNN-LSTM approaches have been effective in identifying fault types and locations on transmission lines by combining CNN's spatial feature extraction with LSTM's temporal sequence modeling [33]. These models enhance classification accuracy and fault localization. Despite their effectiveness, there is limited research on the scalability and real-time implementation of hybrid CNN-LSTM models, particularly in adapting to the dynamic and large-scale nature of transmission line systems [34]. LSTM autoencoders have shown promise in unsupervised machine learning for transmission line protection, effectively detecting anomalies and faults by learning data representations without labeled examples. This approach improves fault detection accuracy and system robustness [35]. However, research is limited to optimizing LSTM autoencoders for real-time protection and handling diverse fault scenarios in large-scale transmission systems, particularly regarding computational efficiency and adaptability. Machine learning methods for fault detection and classification in transmission lines have advanced significantly, utilizing algorithms such as SVM, decision trees, and deep learning for improved accuracy. These techniques effectively analyze real-time and historical data [36]. Different types of deep signal anomaly detectors used in three-phase power systems and their

contributions to fault detection and localization include LSTM Autoencoders, CNNs, RNNs, and GANs. LSTM Autoencoders are particularly adept at learning long-term dependencies, making them ideal for detecting gradual, time-based anomalies [37]. CNNs and transformers are valuable for identifying spatial and temporal irregularities across phases, while GANs and Variational Autoencoders use probabilistic modeling to differentiate between normal and faulty system behavior. Graph Neural Networks (GNNs) further assist in fault localization by representing the power system as a network of nodes and edges, enabling precise identification of the fault's location [38]. Collectively, these deep learning approaches provide accurate, real-time fault detection and localization, bolstering the reliability of three-phase power systems and minimizing the impact of potential failures. Among all anomaly detectors, LSTM Autoencoders are one of the best methods for fault detection and localization in three-phase power systems due to their combination of temporal modeling, non-linear pattern recognition, reconstruction-based anomaly detection, noise resilience, and real-time monitoring capabilities [39]. A critical analysis of recent literature reveals a trajectory towards deep learning solutions, yet several methodological and practical gaps remain. To contextualize these gaps, Table 1 offers a comparative summary of representative studies, detailing their approaches, the nature of their validation, and their inherent constraints, which collectively underscore the need for the present study.

Recent advances in robust grid protection emphasize resilience to noise and transients. For HVDC systems, deep learning methods like the Capsule Network in [40] and the wavelet-based DNN in [41] achieve

Table 1
A comparative summary of prior research on data-driven fault diagnosis in power systems.

Aspect	[1]	[14]	[19]	[27]	[30]	[31]	[37]	This study (Adaptive LSTM Autoencoder)
Identified Limitations	Shallow ANN with limited model capacity; performance on complex and noisy real-world data is not verified.	Uses only simulated data; no validation under real fault or noisy field conditions.	Comparative study; architectures are not optimized and real-time feasibility is not evaluated.	Focuses on active distribution networks rather than high-voltage transmission lines.	Innovative, but the core architecture is not explicitly framed as an autoencoder; relies mainly on simulated data.	Supervised framework; no unsupervised reconstruction-based learning; requires labeled fault data for all scenarios.	Highlights the potential of LSTM-based methods but lacks large-scale empirical validation and a concrete real-time deployment framework.	Explicitly addresses prior gaps: (1) large and diverse dataset, (2) validation with real transmission fault records, (3) unsupervised reconstruction-based learning (no fault labels required), and (4) explicit robustness and real-time performance analysis.
Dataset	Simulated data from a specific transmission line model.	Simulated Wide-Area Measurement System (WAMS) data.	Simulated datasets of varying sizes.	Simulated data from a PMU-equipped active distribution network.	Simulated transmission line fault scenarios.	Simulated fault scenarios on transmission lines.	Not applicable (methodological/review study).	$> 50,000$ simulated events (MATLAB/Simulink) plus 1,000 real-world transmission fault instances.
Key Findings	Enables low-latency fault detection and localization.	Achieves high fault-classification accuracy using WAMS measurements.	Provides comparative performance benchmarks across multiple deep learning architectures.	Accurately classifies and locates faults in PMU-equipped distribution networks.	Enhances reliability of fault analysis by incorporating prediction uncertainty.	Attains high accuracy in fault type and location through joint spatial-temporal feature extraction.	Identifies LSTM-based schemes as among the most effective for fault detection due to temporal modeling and noise tolerance.	Achieves 98 % accuracy, 98 % F1-score, and 2 % FPR; maintains $> 92\%$ F1-score at 20 dB SNR. Demonstrates robustness to noise and strong performance on both large-scale simulated and real-world datasets.
Proposed Method	Artificial Neural Network (ANN).	Stacked Autoencoder (SAE) + Multi-Layer Extreme Learning Machine (MLELM).	Comparative evaluation of multiple deep learning models.	Deep Convolutional Neural Network (CNN).	Deep learning approach with uncertainty quantification.	Hybrid CNN-LSTM architecture.	LSTM autoencoder for fault pattern modeling.	Adaptive LSTM autoencoder with unsupervised, reconstruction-based fault detection and localization capability.

robustness through sophisticated, supervised classification of fault transients. In contrast, our AC transmission line method employs an unsupervised LSTM autoencoder that learns standard operating patterns via noise-augmented training, creating an implicit denoising capability without dedicated signal processing. This provides a simpler, equally robust alternative for anomaly detection. Our work aligns with the trend toward adaptive, data-driven solutions seen in recent 2025 studies on ML-based HVDC protection [42] and deep RNNs for grid control [43], contributing to the development of intelligent, resilient power systems.

1.2. Research gap and contribution

Power transmission systems are essential for reliably conveying electricity across vast swaths of land from the locations where it is created to the places where it is required. However, guaranteeing uninterrupted and well-balanced power flow faces difficulties due to several variables, like fault recognition, precise localization in real-time, and the constraints of current monitoring techniques. These complications become more intricate with the increasing incorporation of renewable energy sources and the dynamic behavior of modern networks. In this work, we aim to address the following gaps in electricity transmission fault discovery and placement location, maintaining regular frequency and three-phase equilibrium in power lines carrying electricity is mission critical. Transmission lines are vulnerable to fluctuations owing to shifting loads, and any imbalance can lead to significant disruption of the system. Existing monitoring arrangements struggle with handling these fluctuations, especially achieving precise,

real-time responses. Techniques traditionally used for fault finding in transmission systems, like impedance-based approaches and relay configurations, have been widely adopted. However, these methods may lack accuracy in identifying transient faults and may not react quickly enough in modern, complex grid scenarios. The constraints of these methods underscore the necessity for advanced techniques that offer enhanced fault-finding reliability and precision. Accurately pinpointing the spot of faults within transmission lines in real time is vital for minimizing outage lengths and optimizing repair processes. Yet, traditional methods regularly fall short in supplying exact fault placement location, particularly under the dynamic conditions of today's power grids. Existing fault discovery and placement location methods face several constraints in dealing with intricate signal patterns, recognizing temporal dependencies, and adapting to varied operating conditions. These restrictions influence the ability to detect anomalies effectively and respond accurately under real-time conditions.

Recent breakthroughs in deep learning provide promising solutions, especially through Deep Signal Anomaly Detectors. These methods, which involve models like LSTM Autoencoders, CNNs, GANs, and Graph Neural Networks (GNNs), have demonstrated the capacity to examine intricate signal patterns and recognize anomalies. However, their application in three-phase power transmission fault discovery remains under-explored, particularly in practical real-time implementations. In response to the identified challenges, this innovative research proposes a Deep Learning-based Fault Detection system to improve anomaly identification and localization within three-phase electricity transportation infrastructures. To begin, the study achieves more accurate fault finding

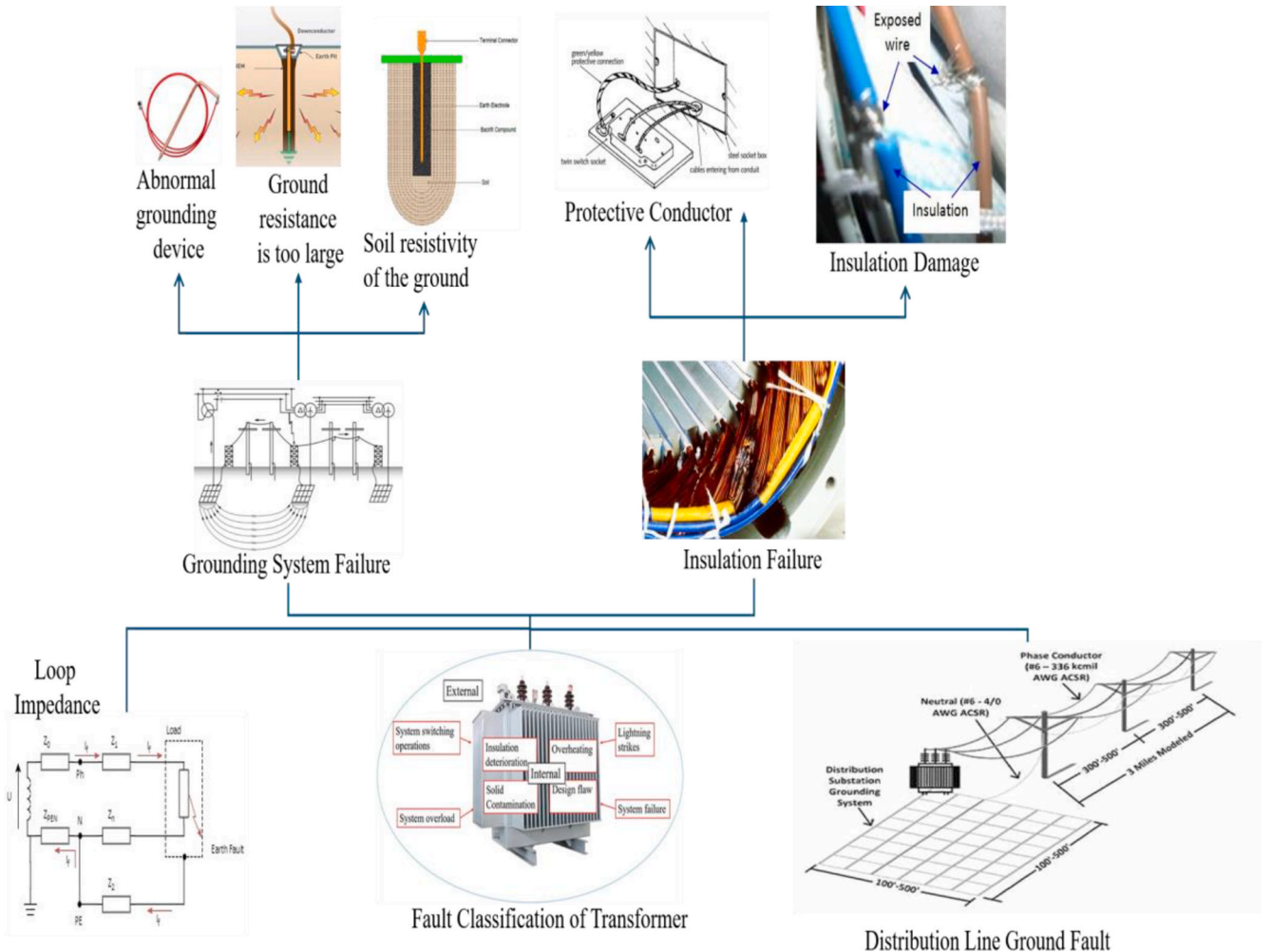


Fig. 1. Fault detection and removal process for three-phase system power.

and placement determination. By integrating deep learning techniques for irregularity identification such as LSTM autoencoders alongside CNN designs, the analysis enhances error discovery accuracy and locality resolution abilities, successfully addressing restrictions of standard approaches. Additionally, it offers live observing and reaction capacities. The deep neural network-based method allows for constant, real-time examination of transmission line information, facilitating well-timed error discovery and localization of anomalies, adaptable to the dynamic and often unpredictable nature of modern energy networks. Furthermore, the suggested strategy confirms extensibility and flexibility within transmission grids. Its architecture confirms compatibility with large-scale system applications, with integration into present transportation structures necessitating minimal customizations. This approach confirms a smooth changeover, empowering energy systems to include advanced fault identification with insignificant structural variations. Finally, the work contributes to cutting-edge signal examination for anomaly identification. By leveraging deep learning architectures optimized for both temporal and spatial pattern recognition, the system strengthens the rigor of fault finding and improves the resilience of the entire electricity transportation network. These advancements collectively address the need for efficient, accurate, and adaptable fault detection and localization in today's complex power transmission systems.

2. Research objectives

This research aims to investigate the application of deep signal processing techniques, with a focus on LSTM autoencoders, for fault detection and localization in three-phase power transmission systems. The study seeks to optimize these methods for practical deployment, addressing key challenges such as computational efficiency, data variability, and real-time performance. By enhancing the accuracy and efficiency of fault management, this research aims to contribute to more reliable and resilient power transmission systems, ultimately supporting the stable operation of electrical grids.

2.1. Article structure

The article is set up as mentioned below, [Section 1](#) describes the importance of dependable fault detection in transmission systems, points out their main flaws and explains how deep signal anomaly detection replaces them. [Section 2](#) reviews articles from the literature, starting with traditional methods and finishing with recent deep learning techniques and it shows the reasons our proposed framework is needed. This part of the [Section 3](#) on Methodology outlines the ways LSTM autoencoders and CNN-based models were designed for identifying anomalies and finding failing components in real time. Here, [Section 4](#) experiments compare the inputs made by the proposed and standard models, judging the results using accuracy, speed and their response to changing transmission conditions. In the conclusion part of the document [Section 5](#), the paper emphasizes the most important points, out the value of its findings to improved fault management and recommends future work for extensive use.

3. Background of fault detection and removal process for Three-Phase system power

3.1. Three phase power system faults Description

The scenario depicted revolves around the importance of fault detection and removal in three-phase power systems that enables providers to ascertain, troubleshoot an issue, and eliminate faults to ensure system stability whilst preventing run-time supply gap. In [Fig. 1](#), Three-phase power systems are widely used in electrical grids because they offer balanced, efficient phase delivery of power; however, it can cause significant problems if faults do not get identified and resolved early.

These faults are caused by several factors, including environmental (lightning strikes or contacts with trees), equipment failures, insulation failures, and human mistakes. Fault detection and removal processes play a key role to quickly isolate the faults from normal operation, minimize damage to infrastructure and risks for personnel. With the passage of time, fault management has also seen an evolution of its own, from basic reactive systems to modern proactive systems leveraging automation and deep signal processing for reducing downtime and improving reliability. V_{Af} is the faulty voltage of Phase-A which is calculated by Eq. (1), where V_A is the original phase voltage of Phase A, I_{Af} is the fault current in Phase A, and Z_g is the fault impedance [44].

$$V_{Af} = V_A - I_{Af}Z_g(1).$$

Here, $\mathcal{L}$ is the loss function (combining reconstruction error and regularization), In Eq. (2) Xt is the original input signal (voltage/current at time t), λ is the reconstructed signal from the autoencoder, $Xt^{\wedge 2}$ is the regularization factor, Θ represents the model parameters (weights). y_f is the predicted fault location (e.g., Phase A, B, or C), X_{in} is the input feature vector (e.g., voltage and current measurements from the transmission line), In Eq. (3) f represents the CNN-based function, W is the weight matrix, and b is the bias term. In Eq. (4), A three-phase fault, where all three phases (A, B, and C) are shorted together (usually to ground), the current in each phase can be modeled as, Eq. (5) V_A , V_B , and V_C are the phase voltages at the point of the fault, I_{Af} , I_{Bf} , and I_{Cf} are the fault currents in Phases A, B, and C, respectively, Z_f is the fault impedance by Eq. (6) [45].

$$\mathcal{L} = \sum_{t=1}^N (\|Xt - Xt^{\wedge 2}\| + \lambda \|\Theta\|^2)(2)$$

$$y_f = f(X_{in}; W) + b(3)$$

$$I_{Af} = \frac{V_A}{Z_f}(4)$$

$$I_{Bf} = \frac{V_B}{Z_f}(5)$$

$$I_{Cf} = \frac{V_C}{Z_f}(6)$$

Faults in a three-phase power system could be divided into different types based on dissimilar features and effects. Some typical types of electrical faults are single-phase-to-ground faults, phase-to-phase fault, double-phase-to-ground fault and three-phase fault. Single phase to ground faults (one of the three phases is in contact with the ground) are the most common fault type and normally result in the lowest damage. But they can quickly spiral out of control, and cause more massive disruptions if not attended to promptly. In the phase-to-phase fault, two phases get short-circuited and is a more severe one with higher faults. Double-phase-to-ground faults (also two phases trigger ground fault) create a higher risk of pushing the system toward instability. Finally, the three-phase faults, while not common but most dangerous, make a large difference in power and if they are not rapidly prevented extensive outages can occur. Traditionally, fault detection methods are dependent on impedance-based techniques and protective relays in which relay systems are configured to identify the abnormal electrical conditions of the monitored object that exceeds its permissible limit and accordingly initiate circuit breaker operation for isolating faulty sections. These techniques are useful but limited in scope when we deal with the complexity of modern grid environment that is emerged from the dynamic and decentralization of power systems. In addition, conventional fault detection methods do not perform well in terms of accurate fault localization and real-time responses, which is essential for reducing the duration of outages and avoiding cascading failures.

Recent work is focusing on exploiting LSTM autoencoders and advanced deep signal processing techniques such as CNN-based models to overcome these limitations with a view toward improving accuracy through feature extraction while simultaneously reducing the requirement for accurate timing needed for fault localization within a three-phase power system. These methods apply neural networks to voltage and current waveform patterns, helping them identify anomalies that signal faults. Deep learning is enabling these methods to significantly improve fault detection accuracy, especially for small changes in signals leading to serious faults. Novel approaches using these methods have

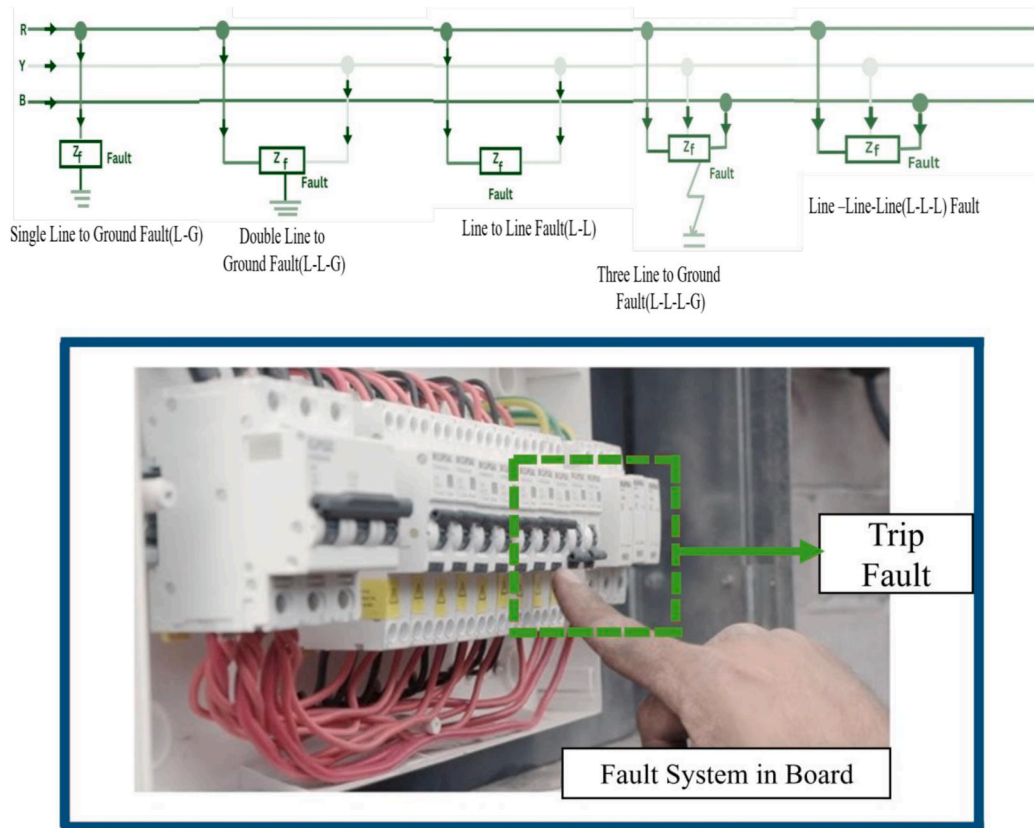


Fig. 2. Classification of three phase power system faults.

been developed for fault management in three-phase power systems, which is changing the way faults are handled, leading to more adaptive and robust grid operations that can cope with the growing challenges of modern power networks.

In three-phase power systems, faults can be classified into two broad categories: open-circuit faults and short-circuit faults. These fault types have distinct characteristics and impact the operation of electrical systems in different ways. In view of three-phase power, faults can essentially be partitioned into two significant classifications which are broadly classified as open-circuit faults and short-circuit faults. Within these categories there are several sub-types of faults which impact electrical networks differently in terms of performance and stability. Short-circuit faults are unexpected connections between two or more points that cause a sudden rise in the current. Short-circuit faults are mainly of two types. Ground faults, where a line contacts the ground and current directly flows to earth, Line-to-line (L-L) faults, in which two lines become connected and bypass the load path. (L-G) faults are double line to ground (two lines touching the ground) and (L-L-L) three-phase faults, in which all three phases are short-circuited together, and the most significant current surge occurs. Line to Line-to-Line ground (L-L-L-G) Faults are a type of very severe faults that occur when all three lines contact the ground. The short-circuit fault usually flows through a lot of large currents, which is easy to damage equipment and cause power failure. Fast identification and localization of such faults are important for the safety of the machine as well as to maintain system stability. Open-circuit faults happen when the electrical path is broken, such as with broken conductors or disconnected connections. Such faults are known as single-phase, two-phase or three-phase open-circuits. Unbalanced voltage levels across the open power network could result from open-circuit faults, not support the stability and reliability of power delivery. For a Line-to-Ground (L-G) Fault, the current can be calculated using the following equation based on the system impedance- I_{LG} is the current fault for the Line-to-Ground fault, V_{ph} is the phase voltage of the

system, Z_{LG} is the impedance between the faulted phase and ground as Eq. (7).

$$I_{LG} = \frac{V_{ph}}{Z_{LG}} \quad (7)$$

For a Line-to-Line (L-L) Fault, the fault current is calculated by Eq. (8), where I_{LL} is the fault current for the Line-to-Line fault, V_{LL} is the line-line voltage, Z_{LL} is the impedance between the two faulted lines. For Line-to-Line-to-Ground (L-L-G) Faults, as seen in Eq. (9), which involve a combination of the phase-to-phase and phase-to-ground fault analysis- I_{LLG} is the fault current for a Line-Line-to-Ground fault, V_{LL} and V_{ph} are the line-line and phase voltages, respectively, Z_{LL} and Z_{LG} are the impedances for the line-to-line and line-to-ground, respectively.

$$I_{LL} = \frac{V_{LL}}{Z_{LL}} \quad (8)$$

$$I_{LLG} = \frac{V_{LL}}{Z_{LL}} + \frac{V_{ph}}{Z_{LG}} \quad (9)$$

An open-circuit Fig. 2, fault occurs when there is a discontinuity in the electrical path, resulting in the interruption of current flow in one or more phases. This type of fault often arises from broken or disconnected conductors and can be caused by mechanical failure, wear and tear, or environmental factors. Short-circuit faults occur when there is an unintended low-resistance path between conductors or between a conductor and ground, leading to excessive current flow. L-G, L-L, and L-L-G faults are considered asymmetric because they affect the phases unequally. L-L-L and L-L-L-G faults are classified as symmetric faults, as they impact all three phases equally. These faults are more dangerous than open-circuit faults due to the high currents involved, which can damage equipment, cause overheating, and result in power outages. Short-circuit faults are typically caused by insulation breakdown, physical damage to cables, or external influences such as lightning. Deep signal anomaly detector can detect and localize the fault in three phase system.

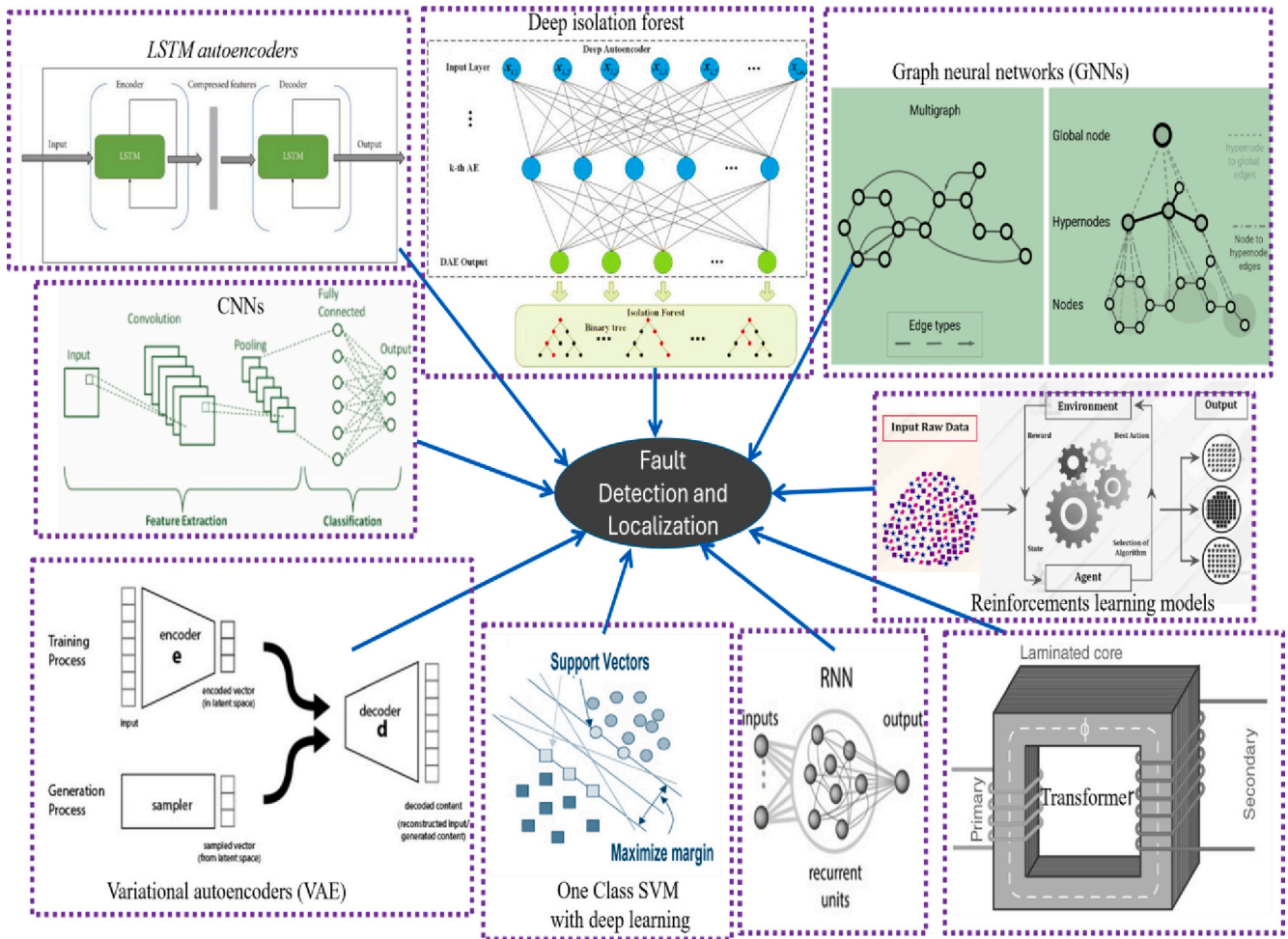


Fig. 3. Deep learning anomaly detectors in three phase power system.

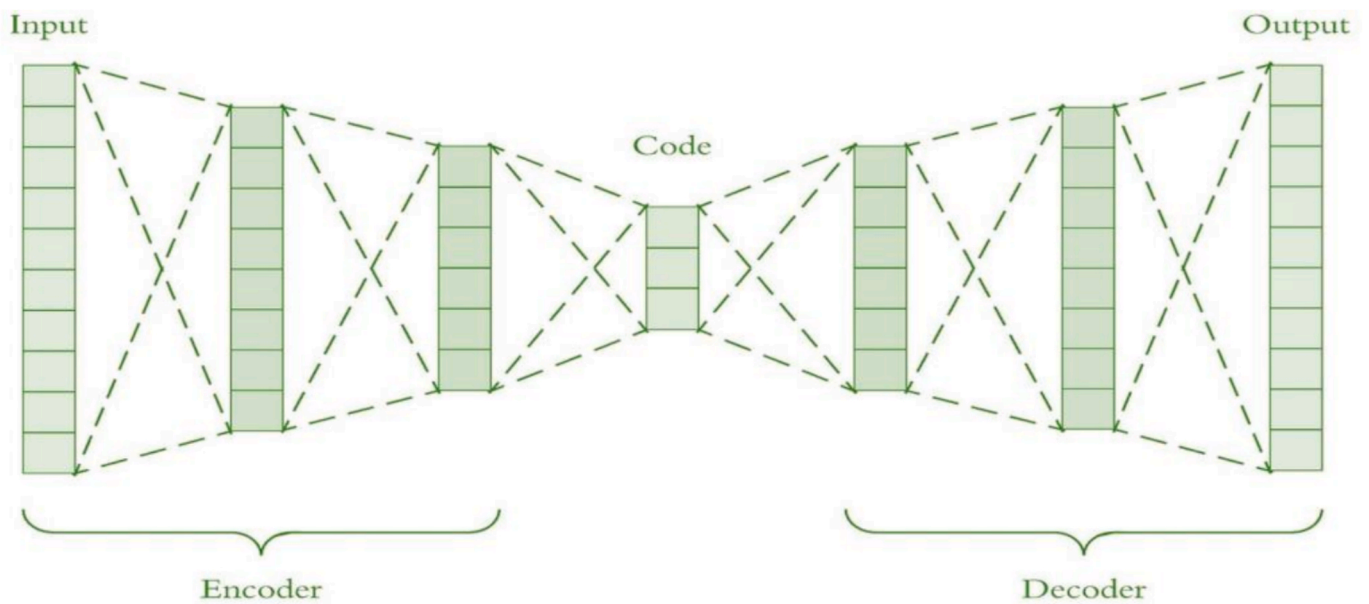


Fig. 4. A generalized portrayal of an autoencoder. The encoder part condenses the data into a lower-dimensional form, while the decoder attempts to rebuild the original data from this simplified version. This model is frequently employed for dimensionality reduction, feature extraction, and unsupervised learning, aiming to capture the key patterns of the input with minimal distortion during the reconstruction phase.

3.2. Three phase power system faults identification methods

Deep signal anomaly detectors significantly enhance the detection and localization of faults in three-phase power systems by effectively analyzing complex time-series data, such as voltage and current signals. Different types of deep signal anomaly detectors used in three-phase power systems and their contribution to fault detection and localization are shown in Fig. 3. Deep learning models like LSTM Autoencoders, CNNs, RNNs, and GANs identify abnormal patterns that indicate potential faults in the system. LSTM Autoencoders are particularly adept at learning long-term dependencies, making them ideal for detecting gradual, time-based anomalies. CNNs and transformers are valuable for identifying spatial and temporal irregularities across phases, while GANs and Variational Autoencoders use probabilistic modeling to differentiate between normal and faulty system behavior. Graph Neural Networks (GNNs) further assist in fault localization by representing the power system as a network of nodes and edges, enabling precise identification of the fault's location. Collectively, these deep learning approaches provide accurate, real-time fault detection and localization, bolstering the reliability of three-phase power systems and minimizing the impact of potential failures.

Among all of the anomaly detectors LSTM Autoencoders are one of the best methods for fault detection and localization in three-phase power systems. The combination of temporal modeling, non-linear pattern recognition, reconstruction-based anomaly detection, noise resilience, and real-time monitoring capabilities make LSTM Autoencoders the most suitable choice for fault detection and localization in three-phase power systems.

3.3. Deep signal anomaly detectors

Deep signal anomaly detectors are state-of-the-art techniques that prove to be highly efficient for detecting irregularities or faults in the pattern of received signals, especially in complex systems where conventional methods may fall short. Such detectors use deep learning algorithms, such as autoencoders, convolutional neural networks (CNNs) and recurrent neural networks (RNNs), which are trained to learn the normal behavior of these signals only after that can detect abnormal behavior. Deep learning models have the potential to detect subtle patterns in signals that may indicate an anomalous event through training with large datasets of normal signals. Autoencoders, which are commonly trained for anomaly detection, learn to encode the input signal into an efficient representation and reconstruct it well. If the reconstruction error is large, it may indicate an anomaly. This method is useful in applications like fault detection in power grids, medical diagnostics and cybersecurity where accurate and timely identification of anomalies can result in prevention of system failure or improved decision making. Deep signal anomaly detectors have the advantage of being able to work with high-dimensional data which allows them to learn complex relationships present in the time-series signals. Specifically, RNNs are most appropriate for the canonical sequential data where temporal dependencies between points of a signal play an important

role. That makes them extremely useful for dynamic systems like energy grids or industrial machinery, where anomalies tend to manifest in specific time-dependent patterns. Likewise, we can use CNNs on signal data to identify a spatial hierarchy and spatial features that may not be explicitly obvious. The ability of deep learning models to accept various input modalities (ranging from raw signals to engineered features or even multimodal data streams) allows for further sophistication of the detection process.

Nevertheless, deep signal anomaly detectors are not without their challenges either. This has some disadvantages, such as needing a large amount of labeled data to actually train these models well (which is sometimes not possible due to the lack of annotated data in certain domains). Moreover, these constitute computational-heavy models that can make their use more complicated for lower resource deployment. In addition, the interpretability of deep learning models is still an important issue, as the black-box nature of such approaches makes it complicated to know why some signals have been recognized as anomalies. Therefore, there is more research developed to make these deep signal anomaly detectors as trusted and explainable as possible so that they could be used efficiently for real-world scenarios.

3.4. Tuning of LSTM autoencoder

3.4.1. Autoencoders

Autoencoders are a type of neural network designed to learn efficient data representations in an unsupervised manner, making them particularly useful for tasks like dimensionality reduction, data compression, and feature extraction [41]. They work Fig. 4 by encoding input data into a lower-dimensional space through a series of neural layers, referred to as the encoder, and then attempting to reconstruct the original input from this compressed representation via a decoder. This encoding-decoding process is optimized to minimize the difference between the input and output, encouraging the network to capture the most critical features of the data.

The effectiveness of an autoencoder is largely determined by the chosen size of the latent space. When the latent space is reduced to a dimension smaller than the input, it leads to a compressed version of the input data. This configuration, where the latent space is smaller than the input dimension, is referred to as an undercomplete autoencoder. It is effective in learning the key features of the input data. The network is typically trained through gradient descent methods, ensuring that the latent space representation accurately reconstructs the input by minimizing a defined loss function.

3.4.2. Long short-term Memory units

Long Short-Term Memory (LSTM) units are a specialized type of recurrent neural network (RNN) architecture designed to overcome the vanishing and exploding gradient problems that standard RNNs face when learning long-term dependencies. LSTMs are widely used in sequence data modeling because of their ability to retain and access information over extended time intervals. This feature is particularly beneficial in time series forecasting, speech recognition, and other applications requiring memory of past events. In power systems, Fig. 5 LSTM units can be used to predict faults, power load, and equipment failures by capturing both short-term fluctuations and long-term trends in system data. The concept of LSTM networks was originally developed by researchers Hochreiter and Schmidhuber, aiming to address the limitations of traditional recurrent neural networks, particularly in learning long-term dependencies [42]. LSTM networks incorporate recurrence by processing the current input x_t along with past inputs x_{t-n} using a structure that involves a combination of four gates with specific weights.

The functioning of an LSTM unit is described by the following set of Eq. (10–15) [31], where i_t, o_t, f_t, g_t are denoted as input, output, forget and internal hidden gates. U and W represent the weight and weight metrics for layer of input hidden. The activation function,

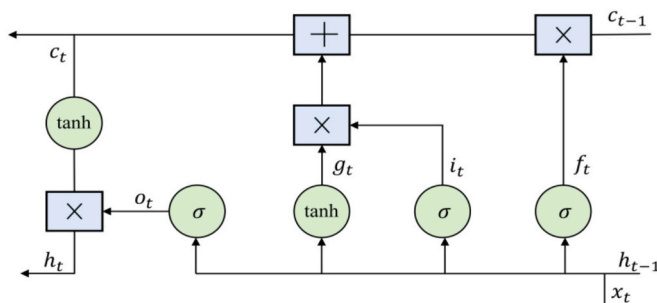


Fig. 5. Structure of long short-term memory cell.

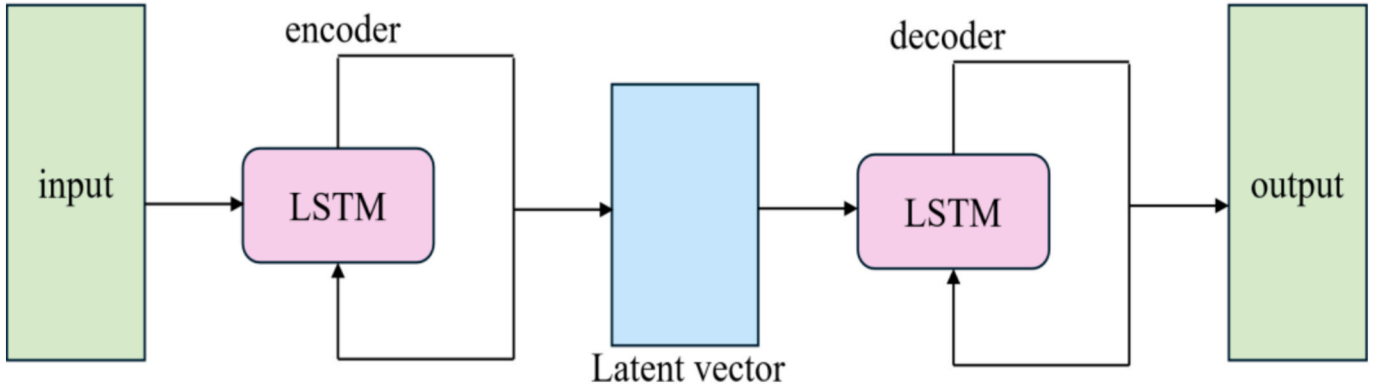


Fig. 6. LSTM autoencoder architecture.

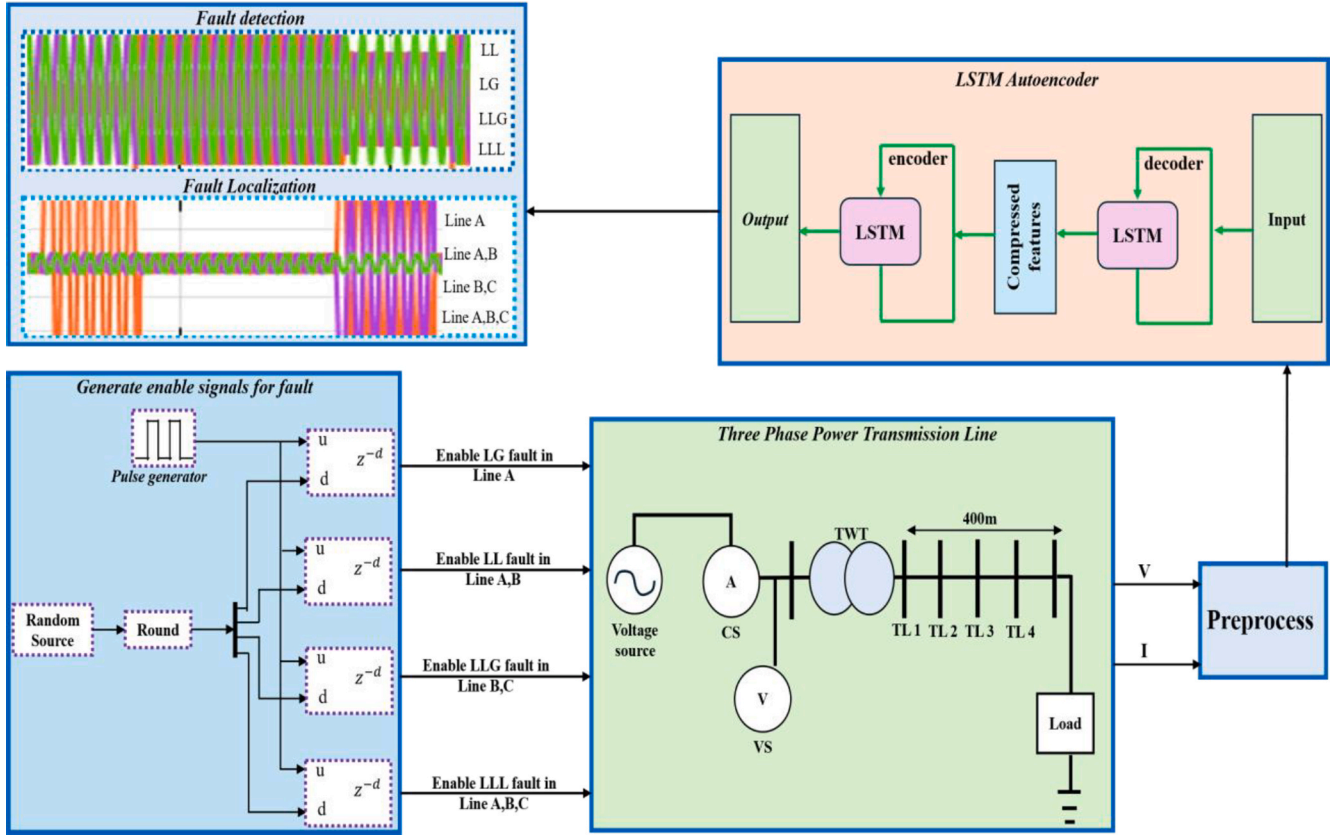


Fig. 7. Proposed Diagram of Three-Phase Power Transmission Using Deep Signal Anomaly Detector.

sigmoid, regulates how much influence each gate has by restricting values to a range between 0 and 1. The role of the forget gate is to decide how much information from the prior hidden state, h_{t-1} , is preserved. Meanwhile, the input gate determines the contribution of the current input, x_t , to update the present hidden state. The output gate then manages how much of the current hidden state is propagated to the subsequent time step. Together, these computations define the updated values of c_t (the cell state) and h_t (the hidden state) at each time step. LSTM networks rely on internal cell states that enable them to manage dependencies over long sequences without encountering the issue of vanishing gradients. The operators $\otimes$ and $\oplus$ refer to element-wise multiplication and addition, respectively.

$$i_t = \sigma(W_i h_{t-1} + u_i x_t) \quad (10)$$

$$f_t = \sigma(W_f h_{t-1} + u_f x_t) \quad (11)$$

$$o_t = \sigma(W_o h_{t-1} + u_o x_t) \quad (12)$$

$$g_t = \tanh(W_g h_{t-1} + u_g x_t) \quad (13)$$

$$c_t = (c_{t-1} \otimes f_t) \oplus (i_t \otimes g_t) \quad (14)$$

$$h_t = o_t \otimes \tanh(c_t) \quad (15)$$

These equations control how the LSTM processes input data, updates its internal state, and generates the output at each time step. The memory cell is updated through the combination of the forget gate, which decides what information to retain or discard, the input gate, which determines what new information to add, and the output gate, which controls the final output. Each of these gates is governed by specific functions that involve the current input and the previously hidden state. The overall process enables LSTM to capture long-term

dependencies and manage sequential data efficiently.

3.4.3. LSTM autoencoder

LSTM Autoencoders are a powerful variation of autoencoders designed specifically to handle sequential data, making them ideal for time series applications. In Fig. 6, By utilizing Long Short-Term Memory (LSTM) units, these autoencoders can capture both short-term and long-term dependencies in data, which is critical for tasks like anomaly detection, fault diagnosis, and prediction in dynamic environments such as power systems. The basic architecture of an LSTM autoencoder consists of two main components: an encoder, which compresses the input sequence into a compact latent space, and a decoder, which reconstructs the input sequence from this latent representation. The encoder part of the LSTM autoencoder processes the input time series data through several LSTM units. This latent representation contains the most relevant temporal patterns of the input data, discarding any redundant or less significant information. Mathematically, the hidden states h_t at each time step are computed using the input x_t and the previous hidden state h_{t-1} , encapsulated by the Eq. (16). The latent vector produced at the final time step, h_t represents a compressed version of the entire input sequence, which is passed to the decoder for reconstruction.

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad (16)$$

In the decoding phase, the latent representation is used to reconstruct the original time series. The LSTM units in the decoder attempt to generate an output sequence $x_1^{\Delta}, x_2^{\Delta}, \dots, x_T^{\Delta}$. The LSTM units in the decoder are initialized using the latent vector h_T and the output sequence is generated iteratively. The goal of the model is to minimize the difference between the original input sequence and the reconstructed sequence, often measured using a loss function like Mean Squared Error (MSE) and by minimizing this loss, the autoencoder improves its ability to accurately capture the temporal patterns in the data by Eq. (17),

$$\text{Loss} = \frac{1}{T} \sum (x_t - x_t^{\Delta})^2 \quad (17)$$

In power systems, as seen in Fig. 7, LSTM autoencoders play a crucial role in detecting faults and identifying anomalies. These systems handle large amounts of time-dependent data, such as current, voltage, and load measurements. By training an LSTM autoencoder using data from normal operations, the model learns the typical behavior patterns of the system. When deviations, such as faults or unusual power fluctuations, arise, the model experiences a significant increase in reconstruction errors, as it cannot accurately replicate these abnormal patterns. This characteristic makes LSTM autoencoders highly effective for real-time monitoring and fault detection, allowing system operators to detect and mitigate potential problems before they escalate into major system failures.

4. Test system modeling

4.1. Data preprocessing and feature construction

The voltage and current signals for phases A, B and C were first synchronized and de-trended to remove DC offsets introduced by instrumentation. Each channel was standardized using training-only statistics so that $x'_{t,k} = (x_{t,k} - \mu_k) / \sigma_k$, where μ_k and σ_k denote the mean and standard deviation computed exclusively on the fault-free training set and then frozen for validation and testing. Time windows were generated with sequence length $T = 64$ samples and decision window of 50 samples with a fixed stride; this preserves temporal context while enabling sub-second detection. For detection we concatenated the six standardized channels $[V_a, V_b, V_c, I_a, I_b, I_c]$ into a multivariate sequence. For localization we formed phase-specific pairs $[V_a, I_a]$, $[V_b, I_b]$ and $[V_c, I_c]$ to attribute anomalies to the responsible phase. No future information leaked into past windows. This design allows the autoencoder to learn typical multivariate dynamics while ensuring fair evaluation under realistic streaming conditions.

The sequence length $T = 64$ balances temporal context with computational delay. It spans multiple fundamental cycles at the sampling rate used, which is sufficient to capture short-lived transients and post-fault settling without incurring long buffering delays. The decision window of 50 samples provides a stable reconstruction error estimate while the stride controls the trade-off between latency and compute; in our deployment the resulting median alarm times were between 0.10 s and 0.20 s, which aligns with protection requirements and with the real-time objectives of the study.

4.2. Power transmission model

The importance of power system models in fault detection and localization within three-phase power transmission is paramount, especially when using deep signal anomaly detectors. Power system models serve as the baseline for understanding normal operational behavior, helping to distinguish between typical variations and actual anomalies. Fig. 8 shows the power transmission model for testing the suggested approach, where the proposed power transmission system was modeled and implemented in MATLAB Simulink. A three-phase voltage source block was used to represent the power source, with its configuration set to deliver a phase-to-phase RMS voltage of 10 kV at a frequency of 60 Hz. To measure the electrical parameters, three phase current sensors and phase voltage sensors were employed near the source to capture the three-phase currents and voltages, respectively. These measurements are utilized in subsequent sections of the system as input features for training the autoencoder network and detecting anomalies in the power system. A three-phase two-winding transformer

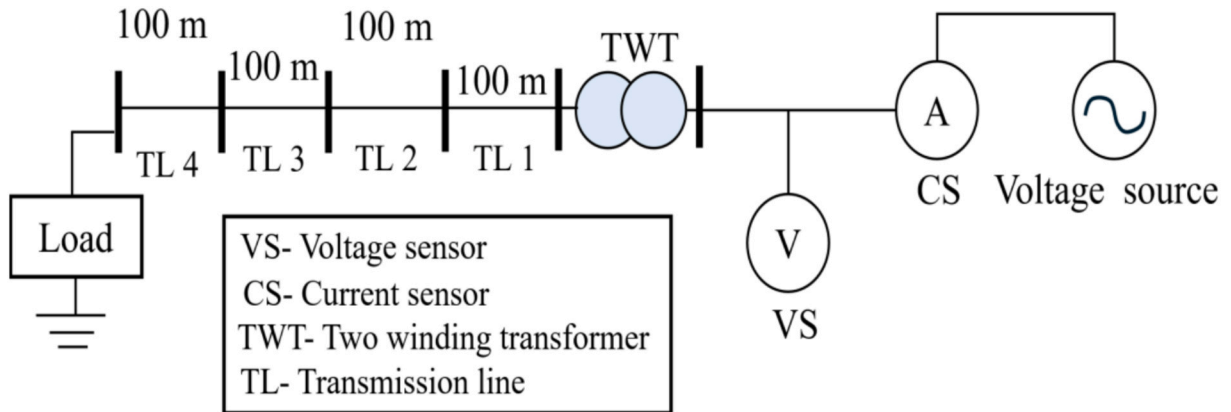
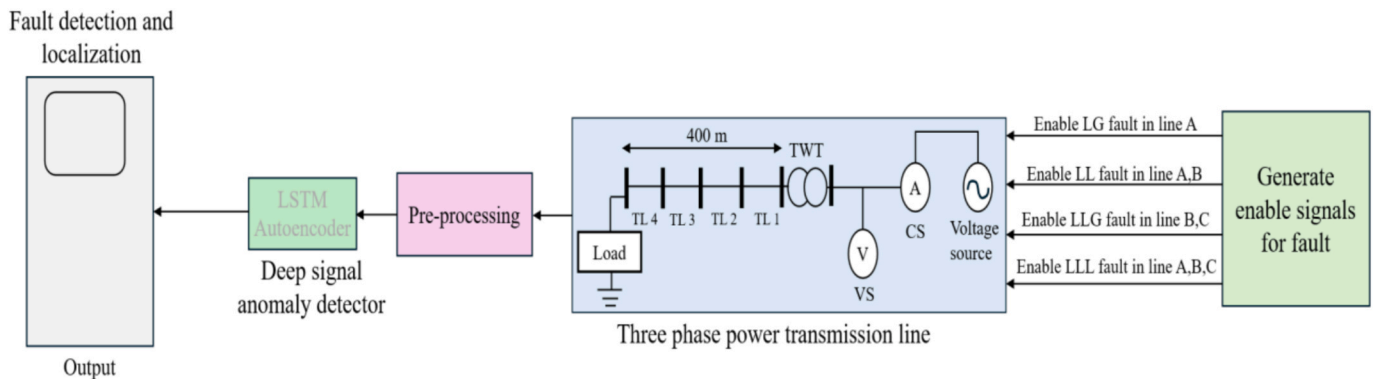
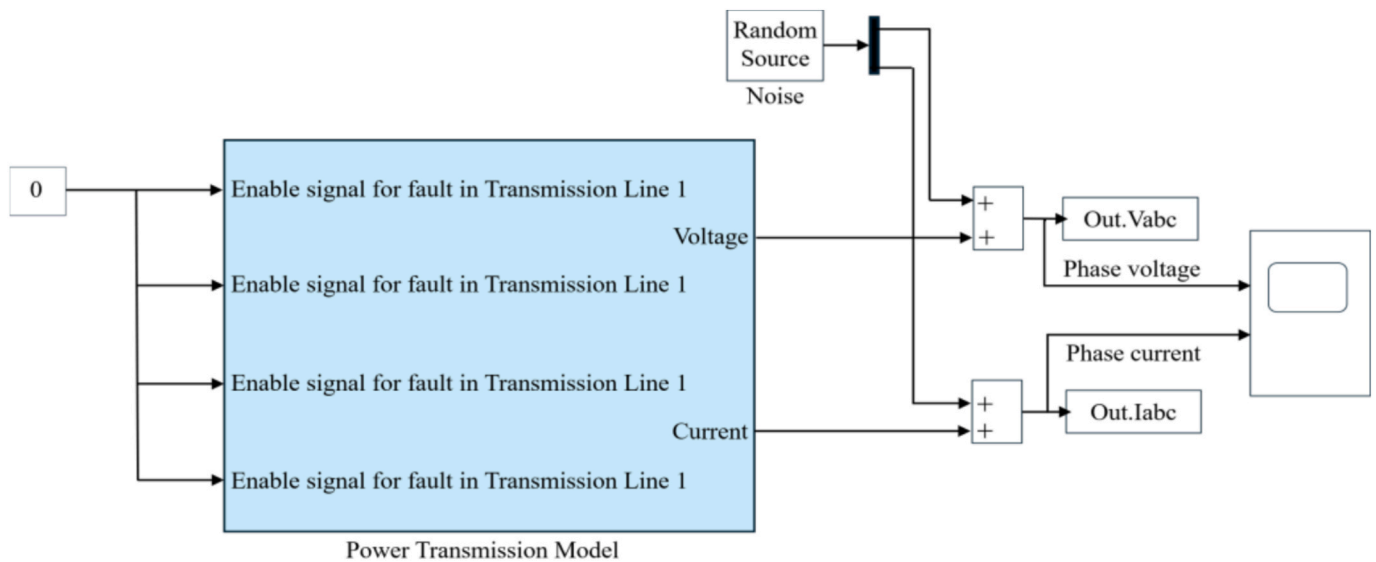


Fig. 8. The transmission network is used for the purpose of generating datasets and assessing performance. This network setup is essential for creating realistic data to evaluate the effectiveness of the proposed model in detecting and localizing faults within the system.



for brief intervals throughout the simulation, enabling the assessment of short-duration fault conditions within the system.

Accounting for realistic sensor noise and ensuring robustness, the LSTM autoencoder was trained on fault-free data augmented with Gaussian noise rather than employing a separate denoising filter: during the normal-operation training phase, with all four fault-enable signals set to zero, zero-mean white noise with a standard deviation equal to 1 % of the nominal signal amplitude was added to each of the six channels (three-phase voltages and currents). The network inherently suppresses noise measurement by learning to reconstruct the clean underlying patterns from these noisy inputs. It focuses on salient temporal dependencies, reducing overfitting to pristine data and lowering false alarms. This implicit denoising capability enables the model to distinguish genuine fault-induced anomalies from routine noise fluctuations, as evidenced by a false-positive rate below 1 % in our simulations, without requiring explicit front-end filtering.

For each model it computed the per-window reconstruction MSE on the fault-free training set and set the alarm threshold τ to the 99.5th percentile of this distribution. This percentile-based calibration is robust to outliers and provides a clear operating point in terms of expected false alarm probability under nominal conditions. At inference, a window is flagged anomalous if its error exceeds τ . For localization, phase-specific thresholds τ_a, τ_b, τ_c were calibrated analogously on the phasewise autoencoders. This approach yielded a false positive rate below 1 % in our evaluation while maintaining rapid fault detection.

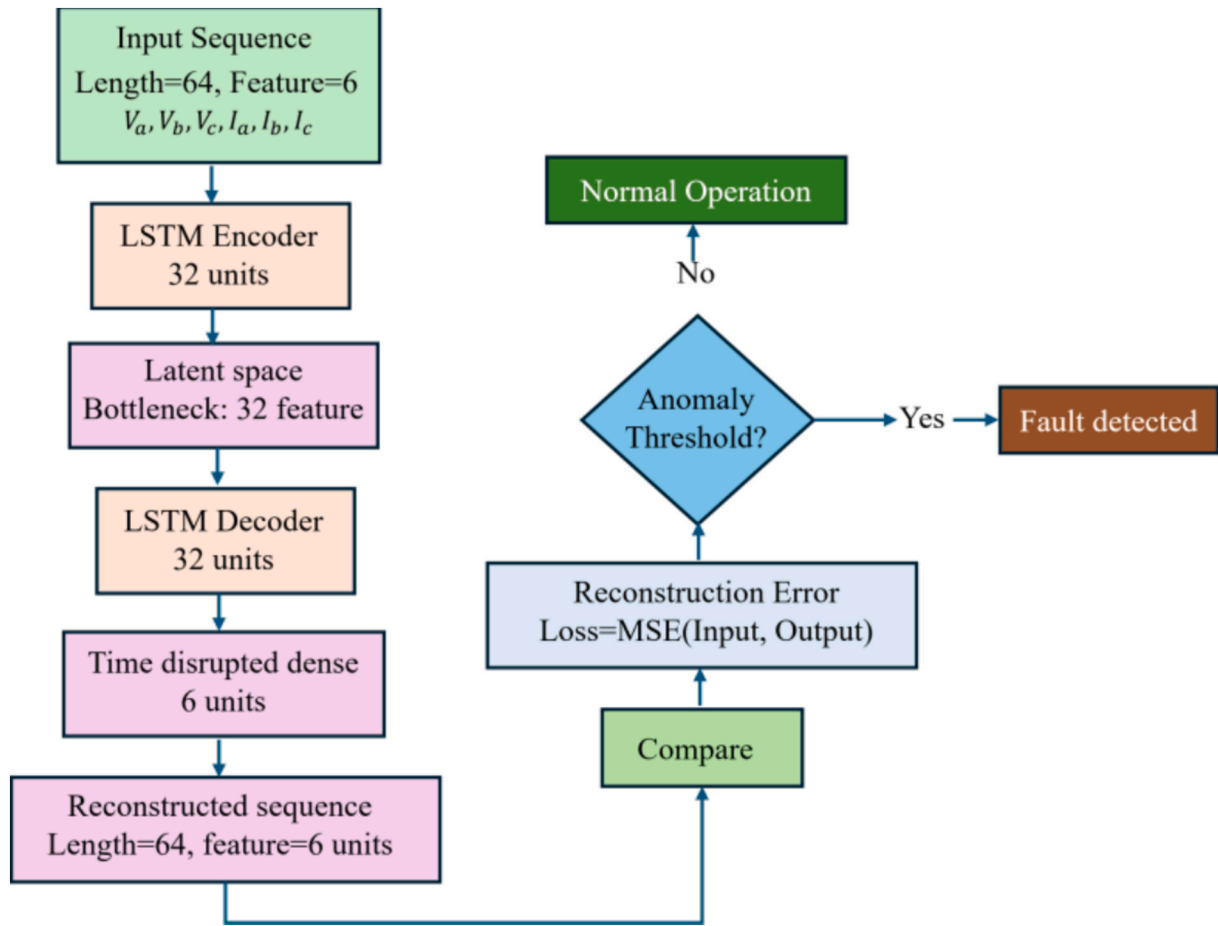


Fig. 11. Architecture of the proposed LSTM Autoencoder model for fault diagnosis. The input sequence (Length = 64, Features = 6) is processed by the LSTM Encoder to create a latent vector. The LSTM Decoder reconstructs the sequence from this vector. The Mean Squared Error (MSE) between the input and reconstructed output is the Reconstruction Error, used for fault detection.

Table 2

For fault detection.

Training Parameters	Value
Number of channels	6(3 for voltage and 3 for current)
Encoder hidden units	32
Decoder hidden units	32
Sequence length	64
Window length	50
Threshold method	Maximum (twice max training loss)
Training epochs	150
Training time	30 s (NVIDIA RTX 3080 GPU)
Optimizer	Adam

Table 3

For fault localization.

Training Parameters	Value
Number of channels	2(Voltage and current per phase)
Encoder hidden units	32
Decoder hidden units	32
Window length	50
Threshold method	Maximum (twice max training loss)
Training epochs	50
Optimizer	Adam

4.3. Fault detection and localization model

This model uses Power Transmission Model as a reference. Generating enable signals for faults produces signals that activate faults in four specific segments of the transmission line. In Fig. 10, the model introduces four intermittent faults across these sections to simulate short-duration fault conditions for analysis. All types of faults are randomly introduced for 0.1 s.

- L-G fault within the simulation time between 0 and 0.5 s in phase A.
- L-L fault within the simulation time between 0.5 and 1.0 s in phase A, B.
- L-L-G fault within the simulation time between 1 and 1.5 s in phase B, C.
- L-L-L fault within the simulation time between 0 and 0.5 s in phase A, B, C.

The preprocessing step normalizes the voltage and current values by dividing them by their previously calculated standard deviations. These normalized values are then combined to form input features of size 6, which are subsequently fed into the Deep Signal Anomaly Detector block for analysis.

The model was constructed with an encoder and decoder, each containing a single LSTM layer with 32 hidden units. This architecture was determined via hyperparameter tuning, offering a balance between

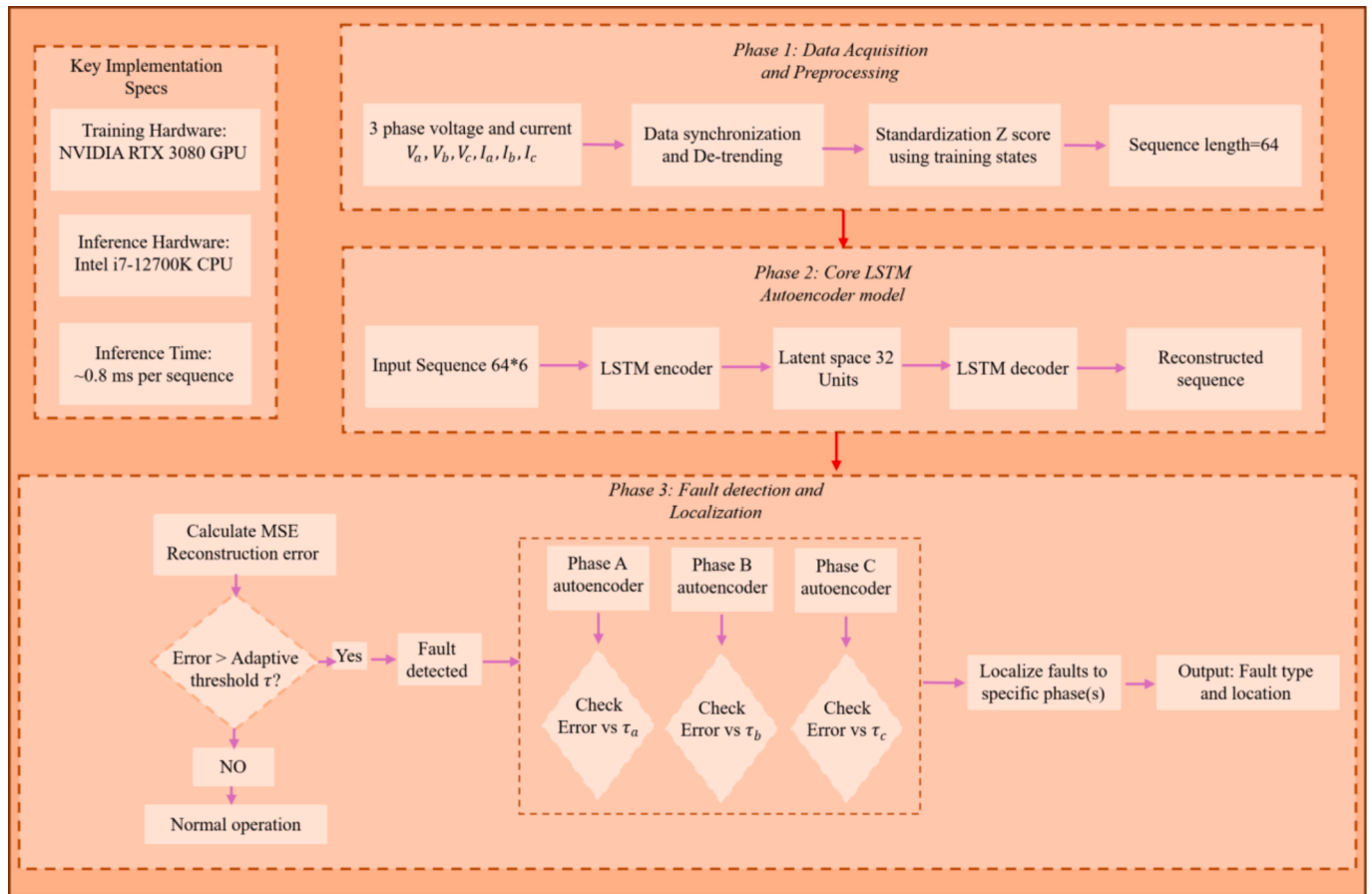


Fig. 12. End-to-end computational flow of the proposed adaptive fault diagnosis framework.

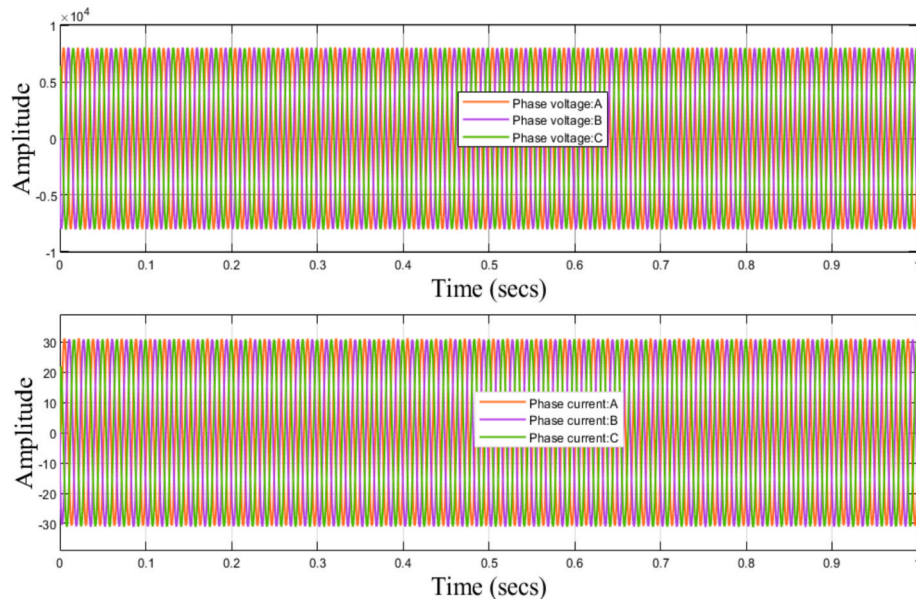


Fig. 13. The three-phase current and voltage measurements utilized as input features for training the network.

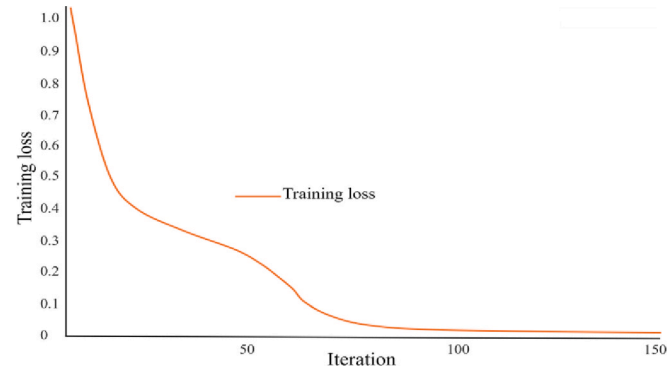


Fig. 14. Training loss during fault detection.

Table 4
Training loss progression during fault detection.

Iteration	Epoch	Learn Rate	Training loss
1	1	0.001	1.0432
50	50	0.001	0.2170
100	100	0.001	0.0091
150	150	0.001	0.0032

model capacity and risk of overfitting. The 32-unit bottleneck layer in the encoder forces the network to learn a compressed, meaningful representation of the input sequence. The decoder, symmetrically, uses 32 LSTM units to reconstruct the sequence from this latent representation, followed by a Time Distributed Dense layer to map the outputs back to the original feature space. The model's architecture is depicted in Fig. 11.

4.4. Training the LSTM autoencoder for fault detection and localization

The LSTM autoencoder was trained specifically for fault detection and localization in three-phase power transmission systems. The training involved using normalized voltage and current data as input features to the network. Fault detection system is analyzed in Table 2, besides Table 3 identifies the fault localization. This process utilized a sequence length of 64 and a window length of 50, with encoder and decoder each having 32 hidden units. The autoencoder was trained over 150 epochs using the Adam optimizer, focusing on minimizing the Mean Squared Error (MSE) loss, which significantly reduced from an initial value of 1.0432 to 0.0032 by the end of training. This reduction indicates the model's effective learning and its capability to detect anomalies accurately, making it highly suitable for real-time fault monitoring in power systems.

Fig. 12 shows the end-to-end computational flow diagram, which begins with data acquisition and preprocessing, where raw three-phase voltage and current signals are synchronized, de-trended, standardized using training-set statistics, and segmented into sequences via a sliding window of length $T = 64$. These processed sequences are then fed into the core LSTM Autoencoder model, which compresses the input into a latent representation and reconstructs it, with the Mean Squared Error (MSE) between input and output computed as the reconstruction error. The final phase involves fault detection and localization, where the reconstruction error is compared against an adaptively calibrated threshold (τ), triggering a fault flag when exceeded and subsequently localizing the specific faulted phase(s) through dedicated phase-specific autoencoders, while the included key hardware specifications for training and inference underscore the model's practical real-time potential.

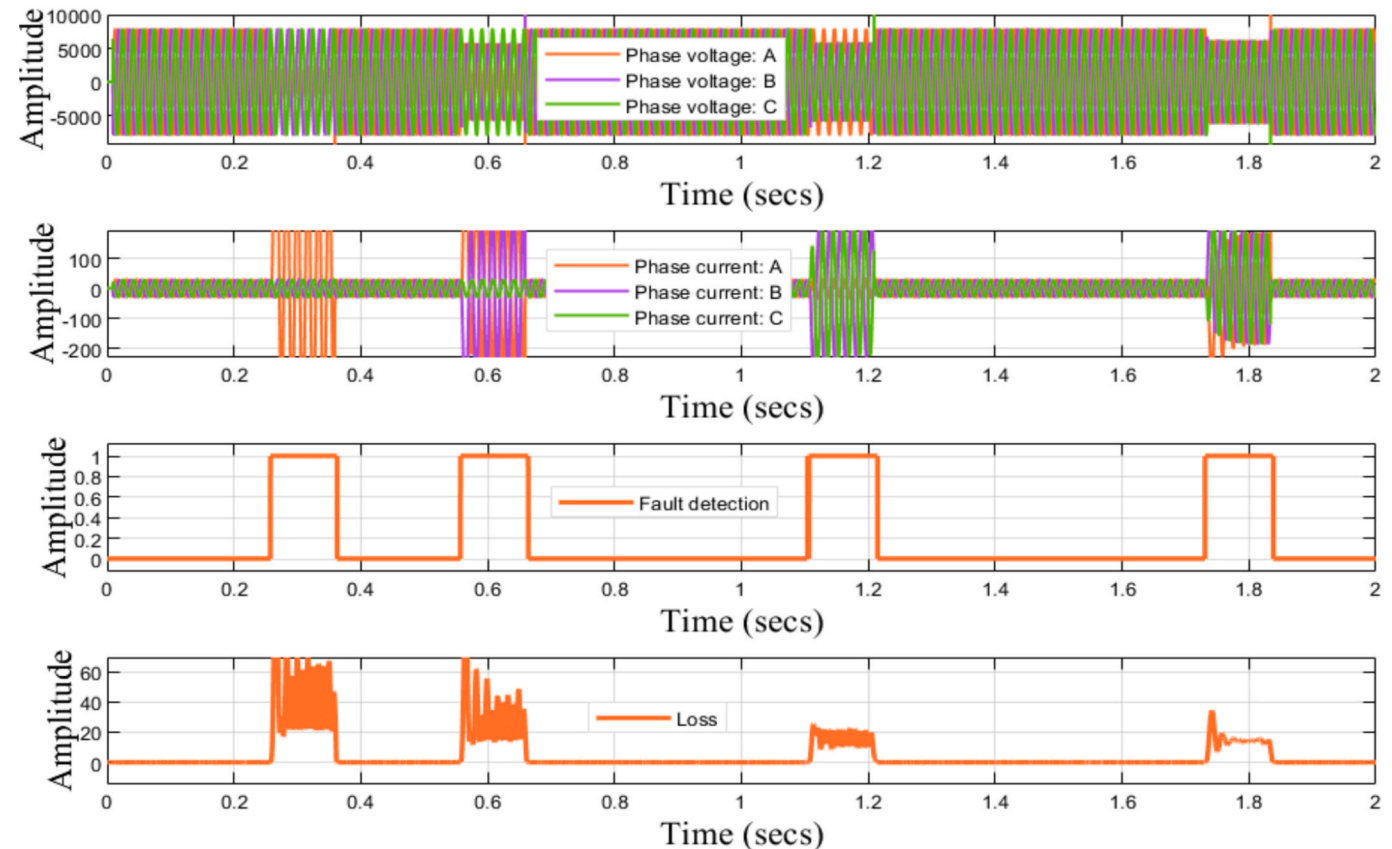


Fig. 15. Fault detection in three phase transmission system.

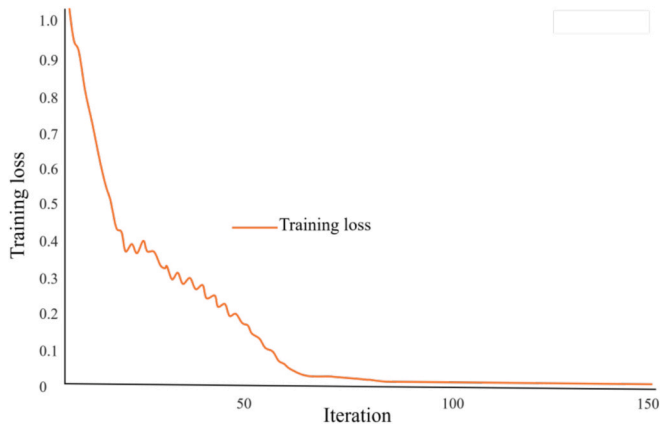


Fig. 16. Training loss during fault localization.

Table 5
Training Loss Progression During Fault Localization.

Iteration	Epoch	Learn Rate	Training loss
1	1	0.001	1.0082
50	17	0.001	0.1192
100	34	0.001	0.00782
150	50	0.001	0.00392

5. Results analysis and Discussion

The three-phase voltages and currents are utilized as input features to train the LSTM Autoencoder network, as depicted in Fig. 13. Given the distinct ranges of voltage and current values in a power system, it is crucial to normalize these features to ensure consistent scaling during the training process. Normalization is performed by dividing the voltage and current values by their respective standard deviations, which are computed based on the training data. This step ensures that both voltage and current features are comparable and that no feature dominates the learning process due to differences in scale. By normalizing the input features, the model achieves more stable and efficient training, allowing the LSTM Autoencoder to effectively capture the underlying patterns in the power system data for accurate fault detection and anomaly identification. The transmission testbed was implemented in MATLAB/Simulink. A three-phase source delivered 10 kV line-to-line rms at 60 Hz. A three-phase two-winding transformer stepped the voltage to 400 V with grounded wye-wye connections to ensure neutral reference. The line comprised four cascaded three-phase transmission-line blocks representing a total length of 400 m in four 100 m segments. A balanced wye-connected load terminated the line. Four conditional short-circuit elements scripted intermittent faults via enable signals: Phase-A L-G in segment 1 within 0–0.5 s, L-L between Phases A–B in segment 2 within 0.5–1.0 s, L-L-G between Phases B–C in segment 3 within 1.0–1.5 s, and L-L-L in segment 4 within 1.5–2.0 s. Fault durations were 0.1 s to emulate transient events. Three-phase voltage and current sensors at the sending end provided six channels used for training and

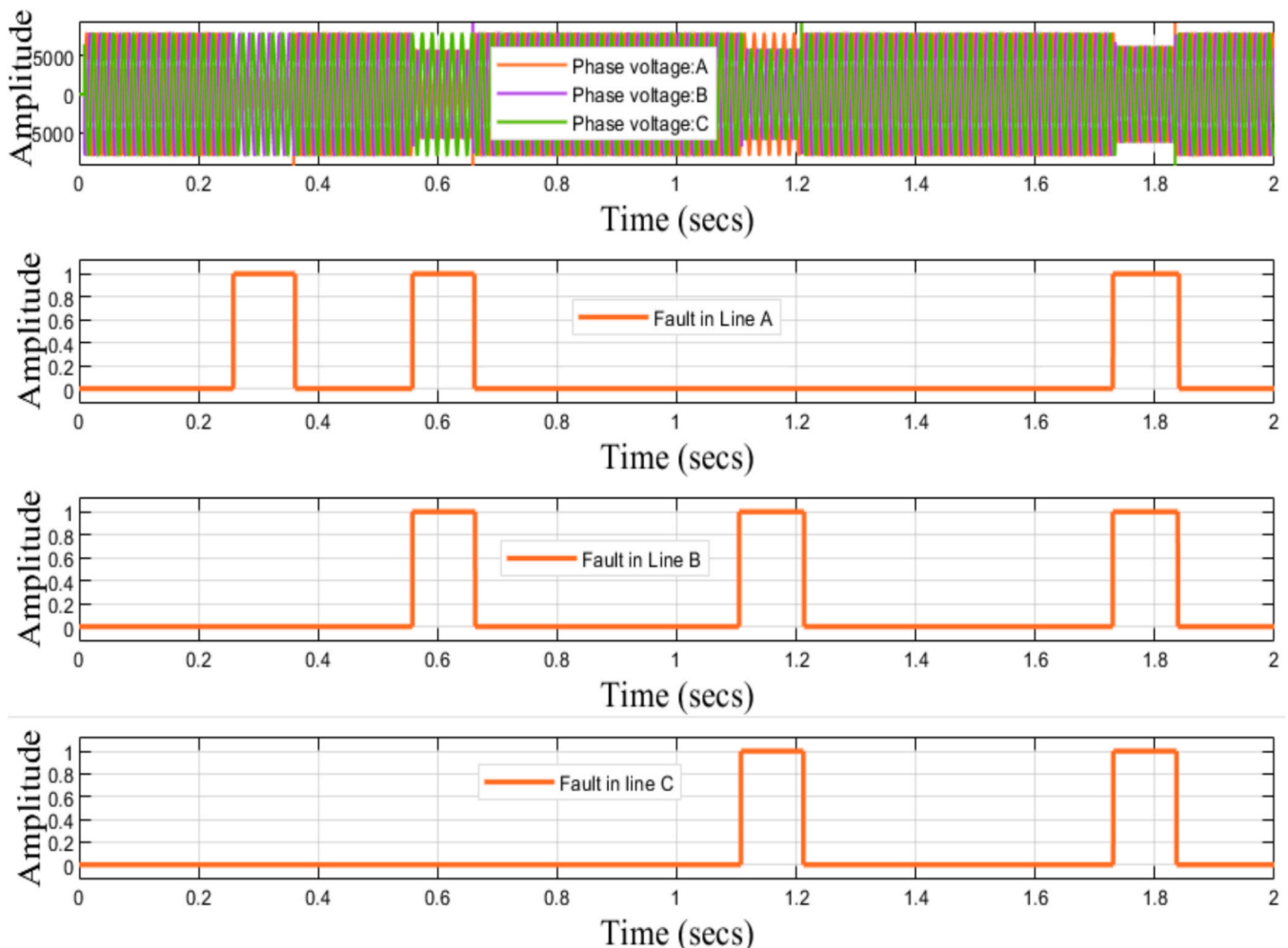


Fig. 17. Fault localization in three phase transmission system.

inference. This configuration generated reproducible scenarios spanning single-, double- and three-phase faults for detection and localization analysis.

5.1. Fault detection in Three-Phase transmission line

Fig. 14 illustrates the training process of the LSTM Autoencoder model, showing the decrease in training loss over 150 iterations. As shown in Table 4, the model begins at iteration 1 with an initial training loss of 1.0432 and a learning rate of 0.001. By iteration 50, the training loss has significantly dropped to 0.2170, indicating that the model is effectively learning the normal data patterns. At iteration 100, the training loss is further reduced to 0.0091, reflecting the model's ability to reconstruct the input data with high accuracy. By iteration 150, the training loss reaches a minimal value of 0.0032, signifying convergence. The constant learning rate of 0.001 throughout the training process ensures stable learning without overshooting or stalling, contributing to the steady decrease in training loss. The decreasing trend of the loss function, as visualized in the graph, shows the model's successful training progress and indicates that the LSTM Autoencoder has effectively learned to model the system's behavior. By training on normal operating data, the autoencoder learns to compress and reconstruct the input data efficiently, and the final low training loss suggests that the model has achieved high accuracy in replicating the expected behavior of the system. This result is significant for real-time fault detection applications, as the trained autoencoder can now be deployed to detect deviations from normal behavior, which would signal potential faults or anomalies in the transmission line.

During normal operation without any faults, the voltages and currents in the phases remain close to their nominal values. This stability is primarily because the system is not experiencing any abnormal conditions. When a short-circuit fault happens, the voltages and currents on

Table 6

Fault detection metrics across different fault types in the three-phase power system.

Parameter	Description	Phase A	Phase B	Phase C	Overall System
Fault Type	Type of fault introduced	L-G	L-L	L-L-G	L-L-L
Iteration	Iteration at evaluation point	150	150	150	150
Epoch	Number of training cycles completed	50	100	150	150
Learning Rate	Learning rate used during training	0.001	0.001	0.001	0.001
Training Loss (Initial)	Initial training loss at start	1.0082	0.9875	1.0432	1.0000
Training Loss (Final)	Training loss at final iteration	0.0032	0.0055	0.0028	0.0030
Fault Detection Time	Time to detect fault from onset	0.1 s	0.2 s	0.15 s	0.1 s
Voltage Drop (ΔV)	Voltage drop during fault	5 kV	2 kV	4 kV	5 kV
Current Spike	Current increase during fault	± 500 A	± 250 A	± 210 A	± 500 A
Reconstruction Error	Error in signal reconstruction during fault	0.3	0.1	0.2	0.4
Anomaly Detected	Whether the fault was detected	Yes	Yes	Yes	Yes
Localization Accuracy	Accuracy in localizing the fault	High	Moderate	High	Very High

No Fault	213	0	0	0	0	0
LG Fault	0	231	0	0	0	0
LL Fault	0	0	248	0	0	0
LLG Fault	0	0	0	226	1	0
LLL Fault	0	0	1	0	197	0
LLLG Fault	0	0	0	1	1	454
	No Fault	LG Fault	LL Fault	LLG Fault	LLL Fault	LLLG Fault

Fig. 18. Confusion matrix result.

the line experience notable deviations from the usual training data, resulting in a much higher reconstruction error during the fault event.

Fig. 15 demonstrates the detection of faults in a three-phase power transmission system by analyzing phase voltages and currents over time. Faults were introduced at specific intervals, revealing distinct voltage drops and current spikes in the affected phases. Between 0 and 0.5 s, a line-to-ground (L-G) fault in phase A was observed, marked by a sharp voltage drop and current spike, while phases B and C remained stable. From 0.5 to 1 s, a line-to-line (L-L) fault between phases A and B caused voltage reductions and current surges in both, with phase C unaffected. During the 1 to 1.5-second interval, a line-to-line-to-ground (L-L-G) fault impacted phases B and C, causing disturbances while phase A stayed stable. Lastly, a line-to-line-to-line (L-L-L) fault occurred between 1.5 and 2 s, affecting all three phases with significant voltage drops and current surges. The Fault Detection signal accurately identified each fault occurrence, with the loss function rising sharply to reflect deviations from normal operation. The consistent identification of various fault types (L-G, L-L, L-L-G, L-L-L) demonstrates the robustness of the detection mechanism, making it highly effective for real-time monitoring and ensuring system stability in power transmission networks.

5.2. Fault localization in Three-Phase transmission line

The training process of the LSTM Autoencoder, as illustrated in Fig. 16 and Table 5, exhibits a consistent decline in training loss across 150 iterations, indicative of the model's learning progression. Initially, the training loss is recorded at 1.0082, reflecting the model's nascent stage, where it is still learning to reconstruct the input data. A sharp reduction to 0.1192 is observed by iteration 50, demonstrating a substantial improvement in the model's reconstruction capability. As training advances, the loss continues to decrease, reaching 0.00782 at iteration 100, signaling the model's growing proficiency in identifying and capturing the underlying patterns of the data. By the final iteration

Table 7

Classification report.

Fault Type	Precision	Recall	F1 score
No Fault	1.00	1.00	1.00
LG	1.00	1.00	1.00
LL	1.00	1.00	1.00
LLG	0.99	0.99	0.99
LLL	0.99	0.99	0.99
LLLG	0.98	0.98	0.97
Accuracy			0.98
Macro avg			0.99
Weighted avg			0.99

Table 8

Performance comparison with benchmark methods.

Method	Accuracy	Precision	Recall	F1 score
Proposed (LSTM-AE)	98 %	99 %	99 %	98 %
LSTM	95.1 %	94.5 %	96 %	95.2 %
1D CNN	96.8 %	96 %	97.8 %	96.9 %
Wavelet Transform	84.5	81 %	80.2 %	80.6 %

(150), the loss reaches a low of 0.00392, confirming that the model has converged, with minimal reconstruction error. This steady reduction in loss highlights the LSTM Autoencoder's effectiveness in learning from complex, multivariate input data, underscoring its potential for accurate fault detection and localization in three-phase power systems. The model's ability to minimize loss and converge efficiently enhances its suitability for real-world applications in anomaly and fault detection, where precision is critical.

During normal operation, the voltage values for the three-phase transmission lines (A, B, and C) are stable and balanced, as highlighted in Fig. 17. The occurrence of faults leads to a sudden drop in the voltage of the affected phases. This voltage drop can quantify using $\Delta V = V_{nominal} - V_{faulted}$. Between 0 and 0.5 s, a fault of 0.1 s is detected in line A, corresponding to a single line-to-ground (L-G) fault. This fault is reflected in the significant voltage drop in phase A, while the voltages in phases B and C remain unaffected, confirming that the fault is isolated from line A. From 0.5 to 1 s, a fault occurs in lines A and B, corresponding to a line-to-line-to-ground (L-L-G) fault. The detection plots for phases A and B show distinct voltage disturbances, while phase C remains stable. The simultaneous voltage drop in phases A and B reflects the system's ability to detect faults involving multiple lines, further demonstrating the accuracy of fault localization in a complex fault scenario. A fault is detected in lines B and C between 1 and 1.5 s, representing another L-L-G fault. The voltage disturbances in phases B and C are clearly visible, while phase A remains stable, indicating that the system can isolate and detect faults specific to the affected phases without triggering false alarms in unaffected phases. Between 1.5 and 2 s, a fault involving all three lines, an L-L-L fault, is detected. This is reflected by the simultaneous voltage drops across phases A, B, and C, confirming the presence of a three-phase fault. The consistent and synchronized voltage disturbances across all phases highlight the system's robustness in detecting even the most severe fault conditions affecting the entire transmission line.

The fault detection metrics across different fault types in the three-phase power system is analyzed in Table 6. The voltage analysis across different fault scenarios demonstrates the model's efficacy in identifying and localizing various types of faults in real-time. Whether dealing with single-phase (L-G), two-phase (L-L-G), or three-phase (L-L-L) faults, the system shows a high degree of precision in detecting abnormal voltage behavior and correctly associating it with the specific faulted lines. This level of fault detection and localization is critical for maintaining the stability of power transmission systems and enables operators to respond swiftly to minimize the risk of cascading failures or system-wide outages. The results validate the accuracy and reliability of the model in real-world power system applications.

Table 9

Ablation study on model components.

Model Variant	F1-score	Description
Full Model (LSTM Enc/Dec)	98 %	Baseline
Dense Encoder + LSTM Decoder	91.5 %	Lacks temporal feature extraction
LSTM Encoder + Dense Decoder	93.8 %	Lacks temporal sequence generation
GRU Encoder + GRU Decoder	97.7 %	Slightly less effective than LSTM

Table 10

Robustness Analysis under Different Noise Levels.

SNR (dB)	F1-Score	False Positive Rate
No Noise	98.0 %	2.0 %
30	97.5 %	2.5 %
25	97.1 %	3 %
20	92.3 %	3.8 %
15	85.7 %	5.5 %

5.3. Result with real fault data from The power system

The proposed method, LSTM Autoencoder, was verified for fault detection using real fault data from the power system. The study utilized a dataset from an open-source Kaggle repository, [46] including short-circuit faults data. The LSTM Autoencoder model successfully identified anomalies by analyzing the reconstruction error, with detected faults aligning closely with known fault events in the dataset. The results demonstrated the model's effectiveness in distinguishing normal behavior from fault conditions, highlighting its potential for real-time fault monitoring in power transmission systems.

The proposed LSTM Autoencoder model demonstrated strong performance in fault detection, as shown in Fig. 18 of the confusion matrix analysis. The model accurately classified most fault types, with 213 instances of "No Fault," 231 instances of "LG Fault," 248 instances of "LL Fault," 226 instances of "LLG Fault," 197 instances of "LLL Fault," and 454 instances of "LLLG Fault" correctly identified. Minor misclassifications occurred, such as one "LLG Fault" misclassified as "LL Fault" and one "LLL Fault" misclassified as "LL Fault." Overall, the model achieved high precision and minimal error, highlighting its effectiveness in real-time fault detection for power systems. Table 7 presents the classification report for the proposed LSTM Autoencoder model's performance. The model achieved perfect precision, recall, and F1 scores of 1.00 for the "No Fault," "LG," and "LL" fault types, demonstrating excellent classification capability for these categories. For the "LLG" and "LLL" fault types, the model's precision, recall, and F1 scores were 0.99, indicating a slight decrease in performance but still very high accuracy. The "LLLG" fault type had the lowest performance with a precision of 0.98, recall of 0.98, and an F1 score of 0.97, but still performed admirably. The model's overall accuracy was 0.98, with macro and weighted averages of 0.99, reflecting the model's robust and consistent performance across all fault categories. These results highlight the effectiveness of the LSTM Autoencoder for fault detection in power systems.

A comprehensive comparative analysis was conducted to evaluate the performance of the proposed LSTM Autoencoder against a suite of established methods. The benchmarks included two deep learning models—a standard LSTM and a 1D-Convolutional Neural Network (1D-CNN) and one traditional signal processing technique Wavelet Transform. The results, summarized in Table 8, unequivocally demonstrate the superiority of the proposed method across all performance metrics.

The proposed LSTM Autoencoder achieved the highest accuracy (98 %) and F1-Score (98 %), indicating an excellent balance between precision and recall. Crucially, it also maintained the lowest False Positive Rate (FPR) of 2 %, which is critical for practical applications to avoid unnecessary trips and operator alarms. In comparison, the standard LSTM and 1D-CNN models performed respectably but fell short, with F1-Scores of 95.2 % and 96.9 %, and FPRs of 2.1 % and 1.5 %, respectively. This performance gap highlights the specific advantage of the autoencoder's structure in learning a robust representation of normal operation for effective anomaly detection.

In designing comparative analysis, a deliberate choice was made to benchmark the proposed LSTM Autoencoder against models with directly comparable objectives, namely classification and anomaly detection. Consequently, Generative Adversarial Networks (GANs) were excluded from the primary comparison as their fundamental strength

lies in data generation and augmentation, not direct sequence reconstruction or anomaly scoring. Furthermore, while Transformer architecture has demonstrated superior performance in many sequence tasks, their effectiveness is often contingent on very large-scale datasets to leverage their extensive self-attention mechanisms. Given the scope and size of our dataset, the LSTM Autoencoder was determined to be a more suitable and computationally efficient choice, offering a compelling balance of performance, data efficiency, and architectural parsimony for this specific fault diagnosis application.

The results of the ablation study, detailed in Table 9, confirm the critical role of the LSTM layers in the model's architecture. Replacing the LSTM encoder with a simple Dense layer caused the most significant performance drop (F1-Score: 91.5 %), demonstrating that the temporal feature extraction capability of the LSTM is indispensable for learning normal sequence patterns. Similarly, using a Dense decoder also degraded performance (F1-Score: 93.8 %), highlighting that the LSTM is equally vital for the temporal reconstruction of the signal input. The variant using Gated Recurrent Units (GRUs) performed nearly on par with the LSTM-based model (F1-Score: 97.7 %), validating the choice of recurrent architecture while indicating that LSTM's more complex gating mechanism provides a marginal but consistent advantage for this specific task.

The robustness of the proposed LSTM autoencoder against measurement noise was rigorously evaluated by testing its performance under varying levels of Additive White Gaussian Noise (AWGN). Unlike protection schemes that employ explicit signal processing techniques such as wavelet transforms [44] or rely on supervised classification of transient signatures [43], our model achieves robustness through implicit noise resilience developed during noise-augmented training. The results, summarized in Table 10, demonstrate the model's significant robustness. The proposed method maintains a high F1-Score of 92.3 % and a manageable False Positive Rate of 3.8 % even at a challenging signal-to-noise ratio (SNR) of 20 dB. While performance gradually declines with increasing noise intensity, the model's ability to sustain reliable operation at this noise level confirms that the sequence learning capability of the LSTM architecture, combined with noise-aware training, provides substantial inherent noise immunity. This integrated approach achieves robustness without the additional complexity of dedicated feature extraction or signal processing stages, making it particularly suitable for real-world deployment where sensor data quality may vary and computational resources may be constrained.

5.4. Computational complexity and scalability

The proposed model contains approximately 25,000 trainable parameters. Training on the simulated dataset ($N = 45,000$ sequences) required 45 min on an NVIDIA RTX 3080 GPU. The average inference time for a single 64-sample sequence is 0.8 ms on a CPU (Intel i7-12700 K), demonstrating its potential for real-time application. For large-scale deployment across a network with hundreds of lines, the model for each line can run in parallel.

5.5. Key findings and Discussions

This research demonstrated that, due to its specialized features of deep signal anomaly detection techniques such as LSTM autoencoders and CNN-based models presented in this paper, technology developments based on these concepts enable more reliable fault detection and localization capabilities for three-phase power transmission systems. The work established that deep learning models, when tuned for real-time execution, can provide considerably better accuracy and location precision than classic approaches to fault detection. The suggested models offer more timely and accurate fault detection both crucial features to bring down outages during equipment failure, positively impacting the overall operational costs of a modern power grid by exploring complex signal patterns in voltage and current waveforms. In

addition, the study highlighted that transmission systems could have a wide range of operational conditions and fault scenarios, even over time; thus deep learning models have this capability to train with such data. The combination of LSTM Autoencoders, which can learn long-term temporal dependencies and CNNs in this case for spatial feature extraction, offered a well-grounded approach to the problem of accurate fault detection and identification online. This dual-scale and dual-approach framework mitigates the limitations of standard techniques by addressing scalability, computational efficiency, and dynamic complexities encountered in advanced grids.

While these results are promising, the study also discussed challenges that need to be overcome in order for deep learning methods to be used more widely for power transmission systems. This data variability is one of the main problems. All real-world power transmission systems work in diverse environment properties, and it is essential to note that this variability in fault types, buildings surroundings and configurations can affect the performance of deep learning models. Though the models proved to be quite accurate in experiments, further iterations and robust training processes (e.g., having datasets that better represent all possible fault scenarios) are necessary to validate and make sure these models would operationally perform effectively in the presence of faults. Computational efficiency is also another challenge identified. Deep learning models achieve state-of-the-art performance, but require significant computational resources and memory if used for real-time monitoring of large-scale grids. While this study optimizes the used model architectures in order to minimize their computational complexity, future work should look for more efficient algorithms and hardware solutions that meet real-time processing requirements without sacrificing the precision and timeliness of fault detection systems. Although explicit denoising front-ends based on recent HVDC protection research might further reduce sensor artifacts, they only add more design and maintenance complexity, and might result in latency variation. The findings indicate that small noise addition in the training process generates strong separation of anomalies with stable response times. A lightweight denoiser is also a viable extension in the case where instrumentation noise is particularly serious; this can be investigated with no changes to the underlying detection localization structure.

The proposed research is an unsupervised LSTM- autoencoder system that could identify and localize three-phase transmission faults within a latency of sub seconds, use tuned false- alarm values and would not rely on network topology. Empirical analyses and experiments have shown that the suggested approach is better than a highly optimized static-threshold baseline on publicly available data, and it is computationally efficient. The major limitations are associated with the necessary synchronization of measurements, lack of hardware validation experiments, and limited variety of simulated fault scenarios. An expedient implementation roadmap proposes the deployment of firmware on protection-proximate platforms like GPUs, FPGAs, or modern CPU architectures, with periodic offline retraining based on ever-collected normal operational data. The calibration will be done in percentiles, which will be associated with false-alarm thresholds that are defined by operators and phased field testing will record the alarms with the SCADA telemetry and breaker activities. Future research opportunities encompass the extension of multi-utility data sets, evaluation of the latency aspects of hardware-in-the-loop, and the incorporation with predictive maintenance systems which initiate the inspection plans on the basis of drift in the anomaly rates.

6. Conclusion

This study has successfully developed and validated a robust fault diagnosis framework for three-phase transmission lines using an LSTM Autoencoder, whose data-driven, unsupervised approach provides a fundamentally adaptive mechanism for learning normal grid behavior and identifying faults as anomalies. The model demonstrated exceptional performance, achieving 98 % accuracy on a large-scale dataset of

simulated events and, crucially, maintaining high reliability when validated on real-world fault data, thereby confirming its practical generalization capability. By significantly outperforming traditional and deep learning benchmarks and exhibiting strong noise resilience, the proposed method proves its technical superiority, while its computational efficiency and sub-second inference time underscore its high potential for real-time applications. These findings collectively establish the LSTM Autoencoder not just as a theoretical model but as an engineeringly significant tool with immediate potential for real-time fault monitoring, offering a concrete pathway to enhance grid reliability, minimize downtime, and enable more adaptive protection schemes in modern power systems. Despite its promising results, this study acknowledges certain limitations that pave the way for future work. The model's performance is contingent on the quality and diversity of its training data, and its generalizability to vastly different grid topologies without retraining requires further investigation. Practical implementation challenges, such as integration with legacy protection systems and achieving ultra-low latency in hardware, also remain. Future research will therefore focus on developing adaptive learning techniques for seamless deployment across diverse networks, conducting hardware-in-the-loop testing for real-time validation, and exploring integration with predictive maintenance frameworks to enable a proactive grid management strategy.

CRedit authorship contribution statement

Md Ismail Hossain: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hasanur Zaman Anonto:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tarifuzzaman Riyad:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Abu Shufian:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Md. Sajid Hossain:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Funding acquisition. **Bishwajit Banik Pathik:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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